

# Multi-source transfer learning method for enhancing the deployment of deep reinforcement learning in multi-zone building HVAC control

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## ABSTRACT

Deep reinforcement learning (DRL) control methods have shown great potential for optimal HVAC control, but they require significant time and data to learn effective policies. By employing transfer learning (TL) with pre-trained models, the need to learn the data from scratch is avoided, saving time and resources. However, there are two main critical issues with this approach: the inappropriate selection of the source domain resulting in worse control performance and inefficient utilization of multi-source domain control experience. To address these challenges, a multi-source transfer learning and deep reinforcement learning (MTL-DRL) integrated framework is proposed for efficient HVAC system control. In order to select appropriate source domains, the contribution of various source domains to the target task is quantified first, followed by a comprehensive evaluation of transfer performance based on average energy consumption and average temperature deviation. The well-pretrained DRL parameters from the optimal multi-source transfer set are then sequentially transferred to the target DRL controller. Results from a series of transfer experiments between buildings with different thermal zones and weather conditions indicate that the MTL-DRL framework significantly reduces the training time of HVAC control, with improvements of up to 20% compared to DRL baseline models trained from scratch. Additionally, the MTL-DRL method leads to reductions in average energy consumption ranging from 1.43% to 3.12% and average temperature deviation up to 14.32%. The impact of the source domain transfer sequence on the performance of the DRL-based control method is also discussed. Overall, the proposed framework presents a promising solution for enhancing DRL-based HVAC control methods by reducing training time and energy consumption while maintaining occupants' comfort.

## 1. Introduction

For people who spend most of their time indoors, the operation of the HVAC system has a significant impact on their comfort, productivity, and health [1]. It is reported that Heating, ventilation, and air conditioning (HVAC) systems consume about 40 %–50 % of total building energy consumption, which imposes a heavy economic burden for building operators [2]. However, inefficient HVAC control methods have been shown to waste 15 %–30 % of building energy [3]. Therefore, the implementation of effective HVAC control methods is essential to reduce energy consumption while ensuring occupants' thermal comfort.

Currently, most HVAC systems are controlled using rule-based and model-based approaches [4,5]. Rule-based approaches require minimal

modeling efforts as they mainly rely on expert rules for control actions. However, such approaches often fail to consider the dynamic nature of HVAC systems, leading to suboptimal performance. Model-based approaches utilize building simulation models to determine optimal control variables. However, developing accurate simulation models for different HVAC systems requires a significant amount of time and professional knowledge. When the model is not accurate enough to describe the thermodynamics of the building system, the control performance may deviate significantly. This represents a serious challenge for the widespread deployment of model-based methods.

Reinforcement learning (RL), particularly deep reinforcement learning (DRL), has gained attention for its model-free and data-driven nature, making it suitable for complex high-dimensional control and optimization tasks in HVAC systems [6]. Researchers have successfully

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| Nomenclature         |                                                                            |                              |                                                                               |
|----------------------|----------------------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------------|
| <b>Abbreviations</b> |                                                                            | $O_{cc,i}(t)$                | Occupancy of zone i at time t                                                 |
| AC                   | Actor-Critic                                                               | $p$                          | State transition probability                                                  |
| DNN                  | deep neural network                                                        | $Q_i(t)$                     | Energy supply of zone i                                                       |
| DPG                  | Deep Policy Gradient                                                       | $r$                          | Reward function                                                               |
| DQN                  | Deep Q-Network                                                             | $\mathcal{R}$                | Total reward                                                                  |
| DRL                  | Deep reinforcement learning                                                | $s$                          | State                                                                         |
| EC                   | Energy consumption, kWh                                                    | $Score(T)$                   | Control performance of the target domain without transfer learning            |
| HVAC                 | Heating, ventilation, and air conditioning                                 | $Score(C_{sj}, T)$           | Control performance after transferring source domain j to the target domain   |
| G1/G2/G3             | Group (1)/Group (2)/Group (3)                                              | $S_j$                        | Source domain j                                                               |
| MDP                  | Markov Decision Process                                                    | $S_{out}(t)$                 | Solar radiation at time t                                                     |
| MSE                  | mean squared error                                                         | $t$                          | Time step                                                                     |
| MTL-DRL              | multi-source transfer learning and deep reinforcement learning             | $T$                          | Target domain                                                                 |
| PPO                  | Proximal Policy Optimization                                               | $T_{avg}$                    | Difference between average temperature of all zones and setpoint temperature  |
| Relu                 | Rectified linear unit                                                      | $T_{in,i}(t)$                | Indoor temperature of zone i at time t                                        |
| RL                   | Reinforcement learning                                                     | $T_{out}(t)$                 | Outdoor temperature at time t                                                 |
| STL-DRL              | single-source transfer learning-deep reinforcement learning                | $T_{sp}$                     | Setpoint temperature                                                          |
| Tanh                 | Hyperbolic tangent function                                                | $T_{th}$                     | Threshold value                                                               |
| TD-error             | temporal difference error                                                  | $\Delta t$                   | Control interval                                                              |
| TL                   | transfer learning                                                          | $ \overline{\Delta T} '$     | Normalized temperature deviation                                              |
| Ts1/Ts2/Ts3          | Transfer scenario 1/Transfer scenario 2/Transfer scenario 3                | $\overline{\Delta T}_{base}$ | Temperature deviation of the target domain without transfer learning          |
| Ts1_v1               | Transfer scenario 1_Reverse order                                          | $\overline{\Delta T}_{T-sj}$ | Temperature deviation after transferring source domain j to the target domain |
| <b>Symbols</b>       |                                                                            | $X$                          | Source network's inputs                                                       |
| $a$                  | Action                                                                     | $Y$                          | Target network's outputs                                                      |
| $b_s$                | Source network's bias                                                      | $\beta$                      | Weight coefficient in the reward function                                     |
| $b_T$                | Target network's bias                                                      | $\theta$                     | Weight and bias parameters of network                                         |
| $c_{penalty,i}$      | Penalty item for indoor temperature deviation of zone i                    | $\mu_{T,i}$                  | Average temperature deviation of zone i                                       |
| $EC'$                | Normalized energy consumption                                              | $\pi$                        | Policy                                                                        |
| $EC_{base}$          | Energy consumption of the target domain without transfer learning          | $\bar{\sigma}$               | Average of the standard deviation of temperature deviation across all zones   |
| $EC_{T-sj}$          | Energy consumption after transferring source domain j to the target domain | $\sigma_{T,i}$               | Standard deviation of temperature deviation of zone i                         |
| $L$                  | The number of hours per week during the testing process                    | $\omega_s$                   | Source network's weights                                                      |
| $N$                  | The number of zones                                                        | $\omega_T$                   | Target network's weights                                                      |

applied DRL to HVAC scheduling, radiant heating system regulation, energy consumption optimization, and supply temperature control [7–10]. However, the learning process in DRL control methods often requires extensive exploration and training periods to achieve desired proficiency, especially in complex system architectures. Moreover, the generalizability of DRL methods may be limited, necessitating repeated training for specific building configurations.

Transfer learning (TL) offers a potential solution to enhance DRL control methods in HVAC systems by reducing training time and improving efficiency. The main idea is to leverage the transferability advantage of neural networks to transfer knowledge learned from a source task to another different but related target task [11]. Transfer learning has been widely applied in various fields with positive effects, including building energy prediction [12], thermal comfort prediction [13], and HVAC fault detection [14]. Consequently, the integration of TL and DRL control methods holds promise for enhancing training efficiency and control performance. However, this is not a simple process. A crucial prerequisite for the successful application of transfer learning is the selection of the source domain. An inappropriate source domain could adversely affect the effectiveness of transfer learning [11]. As each building is personalized [15], with different sizes, layouts, the number of thermal zones, and weather conditions, knowledge transfer between different buildings may be ineffective [16]. Therefore, selecting a source

domain that is conducive to the target task is essential for leveraging the benefits of transfer learning. Additionally, there are limitations to address when applying transfer learning to DRL-based HVAC control methods. Many studies have primarily focused on transferring HVAC control strategies from a single source task to another target task. However, there are often multiple training scenarios related to the target task, overlooking potential information from other source domains due to multiple training scenarios associated with the target task.

This study aims to address two key challenges in utilizing transfer learning within DRL-based HVAC system control methods. The first challenge is the selection of the source domain. Previous studies typically evaluate the similarity between the source and target domains based on key parameters. However, this may increase the risk of selecting an inappropriate source domain, resulting in suboptimal performance after transfer, as the control performance of HVAC systems is influenced by the complex relationship of various parameters. The second challenge involves effectively leveraging control experience from multiple source domains. In practical scenarios, there are often several training scenarios relevant to the target task. However, many studies only transfer knowledge from a single source domain, overlooking valuable information from other domains.

To solve these challenges, we develop a multi-source transfer learning and deep reinforcement learning (MTL-DRL) integrated control

framework for multi-zone buildings. This framework selects appropriate source domains to construct the optimal multi-source transfer set by comprehensively evaluating the contribution to the target task and exploring the transferability of DRL-based HVAC system control strategies. Firstly, the contribution of each source domain is quantified based on the control performance of energy consumption and zone temperature. The transfer performance of each source domain is comprehensively evaluated in terms of average energy consumption and average temperature deviation to select the source domain with optimal transfer performance. Then, pre-trained source parameters from the optimal source domain set are sequentially transferred to optimize the target DRL controller through learning from various source domains. We conducted various transfer experiments between buildings with different numbers of thermal zones and weather conditions to demonstrate the effectiveness of our approach in reducing training time, and energy consumption with minimal temperature deviations. Additionally, we analyze the impact of the source domain transfer sequence on the performance of DRL-based HVAC system control.

## 2. Literature review

The design methods for HVAC control systems can be broadly classified into model-based and data-driven approaches. Model-based methods necessitate precise thermodynamic models for specific buildings to achieve accurate results [5,17], which presents challenges for their widespread implementation. Data-driven methods address the limitations of model-based methods, as they do not rely on explicit physical models. These methods often employ machine learning techniques, specifically deep reinforcement learning (DRL), which is well-suited for controlling HVAC systems with the nonlinear, time-varying, and complex nature [18]. Many studies have applied DRL techniques to various aspects of HVAC system control, such as the entire HVAC system [19,20], and associated energy management systems [21–23]. For instance, Barrett and Linder [24] utilized a DRL approach to improve thermal comfort and reduce energy consumption in residential buildings, achieving up to 10 % energy savings compared to programmable thermostats. Fang et al. [16] proposed a DRL-based framework for building HVAC system control. Their results demonstrated that the DRL control strategy can reduce the energy consumption of the HVAC system of 14 fixed temperature setpoint comparison cases, while maintaining acceptable indoor air temperature deviation. These advanced DRL-based control methods show significant potential in enhancing indoor comfort conditions and reducing energy consumption.

Some studies have explored online DRL controllers that learn optimal control strategies through real-world interactions [22,25]. However, the online method requires extensive training time to interact with the environment to achieve near-optimal control strategies. In [25], an online DRL method learned the optimal schedule for the energy management system in one year. Therefore, the commonly adopted strategy in the current literature is offline pre-training before deploying the DRL controller. Once the DRL controller has received sufficient training on a building surrogate model, it can be deployed in the real world. However, the offline pre-training of DRL controllers is limited to specific building configurations, resulting in poor scalability and generalizability of these control strategies. To address this limitation, transfer learning (TL) emerges as a promising technique that enables the utilization of pre-acquired knowledge in different but related tasks and improves the scalability of DRL-based building controllers [11]. TL techniques facilitate information transfer between controllers, simplify algorithm deployment for specific control problems, and enable RL-based controllers to exhibit acceptable performance from the initial deployment stages. However, the application of TL in building system control is limited compared to other fields related to RL-based agents like robotics [26].

In [27], the effectiveness of reinforcement learning in pre-training and fine-tuning neural networks for HVAC control is evaluated. First,

the RL agent is trained in a building simulation environment to establish a baseline model, followed by fine-tuning in two independent simulation environments. One environment simulates different weather conditions for the same building, while the other simulates the same weather conditions for a different building. Results indicate that transfer learning agents outperformed rule-based controllers, improving by 1 % to 4 % compared to agents trained from scratch. Xu et al. [28] designed a novel transfer learning method that utilizes the front network layers of a neural network controller to capture building-agnostic behavior, while the last network layers were trained for each specific building. Their neural network controller comprises a transferable front-end network and a building-specific back-end network. Through simplified multi-zone building models, they evaluated the performance of transfer learning when the RL control strategy was transferred to the target building under varying weather conditions, envelope features, number of thermal zones, and HVAC configurations. Results demonstrate the successful transfer of the Deep RL controller from a source building to a target building. Similarly, Fang et al. [16] discussed the impact of the number of neural network layers on the knowledge sharing process. Their proposed method also showed that the transfer learning had better performance when DRL agents were transferred between buildings located in similar weather conditions. Transfer learning in [29] was utilized to transfer the pre-trained optimal strategy from the source domain to each zone of the target building for VAV damper position management, resulting in a 40 % reduction in HVAC energy consumption compared to the baseline controller and a 50 % decrease compared to RL agents trained from scratch over 5,000 months. Lissa et al. [30] introduced a TL-based methodology to facilitate the sharing of RL control strategies among HVAC systems with different spatial and geographical characteristics. Their findings indicate that in the absence of TL, the proportion of daily occupant discomfort time ranges from 7 % to 36 %. Conversely, the implementation of TL decreased this proportion to 3 %, thereby enhancing indoor comfort conditions. Additionally, they also report that TL accelerates optimal convergence in the presence of geographical changes. Zhang et al. [31] investigated the transfer of RL agents to homes with similar appliances but different user preferences and parameters, highlighting the effectiveness of TL in reducing training time for the RL controller based on the similarity of homes. Mbuwir et al. [32] evaluated the potential of transfer learning to transfer battery management control policies between similar buildings, which is identified through energy consumption patterns using the K-shape clustering algorithm. Simulation results show that transfer learning does not only speed up the convergence of the learning algorithm but also improves its scalability by sharing knowledge between buildings with similar characteristics. The authors in [33] enhanced the learning process of the model-free HVAC controller by combining TL techniques with unsupervised learning to match households with similar occupancy patterns. Compared with the conventional model-free controller, their approach reduced the deviation period and average temperature deviation by approximately 4 % and 68 %, respectively.

In summary, previous studies usually evaluate the similarity between the source and target domains by investigating a limited number of parameters, as shown in Table 1. However, the selected source domain may not improve the performance of the target task after transfer. As the similarity between the source and target domains is often influenced by the complex relationship of various parameters including occupancy, energy consumption, outdoor weather conditions, and HVAC characteristics, it makes it more challenging to identify the optimal source domain based on a few parameters. Therefore, our approach focuses on investigating the transfer contribution of the source domain to the target task. We identify the source domain with the most significant contribution and construct the optimal source domain set. Then, the pre-trained source parameters from the optimal source domain set are transferred to the target DRL controller in sequence. Our unique source domain selection method forms the foundation of the proposed multi-source transfer framework.

**Table 1**  
Summary of studies applying TL in DRL-based building HVAC system control.

| Ref.      | Building type | Source and Target domains | Selection criteria of source domain | Number of source domains |
|-----------|---------------|---------------------------|-------------------------------------|--------------------------|
| [16]      | Multi-zone    | Same                      |                                     | Single                   |
| [27]      | Multi-zone    | Different                 |                                     | Single                   |
| [28]      | Multi-zone    | Different                 |                                     | Single                   |
| [29]      | Multi-zone    | Different                 |                                     | Single                   |
| [30]      | Single-zone   | Different                 |                                     | Single                   |
| [31]      | Single-zone   | Same                      | Smart home                          | Single                   |
| [32]      | Multi-zone    | Different                 | Energy consumption                  | Single                   |
| [33]      | Multi-zone    | Same                      | Occupancy                           | Single                   |
| This work | Multi-zone    | Different                 | Contribution on target task         | Multiple                 |

### 3. Methodology

#### 3.1. Overall flowchart of this study

The proposed framework in this study mainly consists of four parts, illustrated in Fig. 1. The first step is to formulate the HVAC control strategy, which mainly defines the state, control actions and reward function of the DRL-based control method. The details of this step are described in Section 3.2. The second step is the selection of the source domain. In this study, we developed an evaluation framework to quantify the transfer contribution of source domains to the target domain and construct an optimal multi-source transfer set. Subsequently, the multi-source transfer learning and deep reinforcement learning (MTL-DRL) integrated framework was designed. The experiences of well-trained control agents were sequentially transferred to the target agent in order to accelerate the learning process and improve control performance. Detailed explanations are provided in Sections 3.4 and 3.5. In order to evaluate the transferability of the MTL-DRL method, a series of transfer scenarios were conducted between buildings with different numbers of thermal zones and weather conditions. The transfer performances are discussed in Sections 4.4 and 4.5.

#### 3.2. Mapping multi-zone HVAC control problem to Markov decision process (MDP)

This section formulates the control problem of multi-zone building HVAC systems as a Markov decision process (MDP). Based on the simplified thermal dynamics model of HVAC systems in [34], the indoor temperature at the current time interval is only related to the previous state parameters, such as the indoor temperature at the previous time interval and is not affected by the indoor temperature at any other time intervals. Therefore, the control problem of HVAC systems can be regarded as a finite Markov process and be solved using the DRL method. The DRL based control model for building HVAC systems consists of

environment, agent, states (s), actions (a), and reward function (r). The building HVAC system can be considered as the environment, and the DRL control strategy can be regarded as the control agent. The agent perceives the environment through the state space, which provides relevant information about the present situation. Then the agent selects an action from its available set of actions, which represents how the agent interacts with the environment. As the environment transitions to a new state in the next step, the agent receives an immediate reward that quantifies the desirability of the action taken. The reward function provides guidance for the learning process of the agent to enable better decision-making over time. The state (s), action space (a), and reward function (r) are designed as follows:

**State space.** The state space  $s_t$  represents the current environment information required for the decision-making of the DRL control agent. The indoor environment state parameters include the temperature and occupancy of each thermal zone, denoted as  $\{T_{in,1}(t), T_{in,2}(t), \dots, T_{in,n}(t)\}$  and  $\{O_{cc,1}(t), O_{cc,2}(t), \dots, O_{cc,n}(t)\}$ , respectively. The outdoor environment state parameters include the outdoor temperature  $T_{out}(t)$  and solar radiation  $S_{out}(t)$ , where  $t$  represents the current time. Therefore, the state space  $s_t$  can be designed as:

$$s_t = (T_{in,1}(t), T_{in,2}(t), \dots, T_{in,n}(t), O_{cc,1}(t), O_{cc,2}(t), \dots, O_{cc,n}(t), T_{out}(t), S_{out}(t)) \quad (1)$$

**Action space.** The action space  $a_t$  is a set of continuous controllable actions formed by the energy supply to the HVAC system. Each zone of the HVAC system is equipped with a thermostat. When the indoor temperature is higher than the set value, the HVAC system will turn on to cool the room; otherwise, it will provide heat to keep the occupants comfortable. The whole action space is limited by the maximum energy supply, which is set at 150 kWh. The action space  $a_t$  is defined as follows:

$$a_t = (Q_1(t), Q_2(t), \dots, Q_n(t)) \quad (2)$$

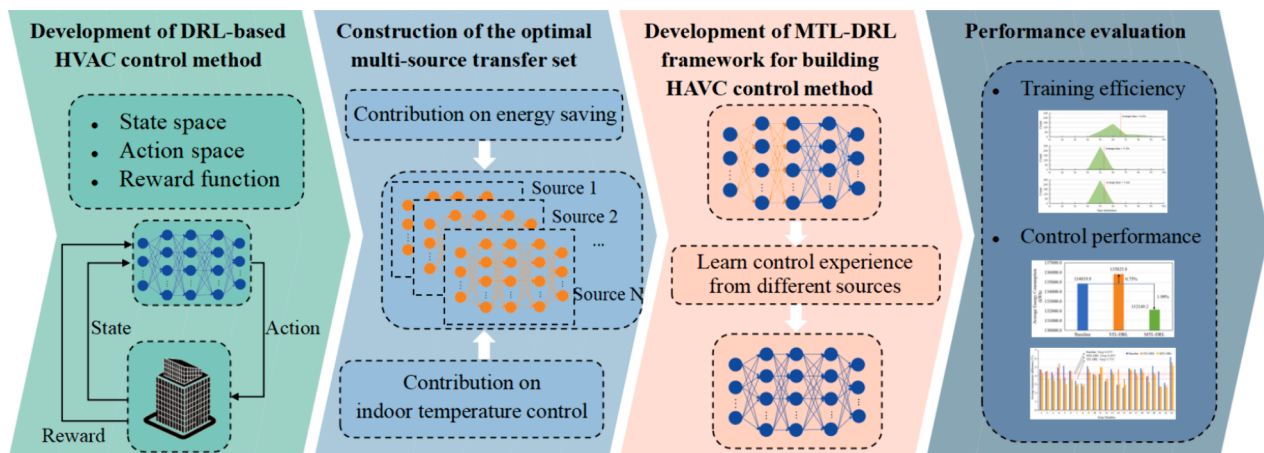


Fig. 1. Overall flowchart of this study.

**Reward function.** The objective of building HVAC system control is to maintain indoor temperature within given comfort range while minimizing energy consumption. The energy-related term represents the total energy consumption of the HVAC system, while the temperature-related term is the cumulative violations of indoor temperature deviations from the temperature setpoint. The reward function  $r(t)$  acts as a penalty, more energy and temperature penalties will be added to the reward function when inappropriate control actions are taken, and the state deviates from the acceptable range. The reward function can be designed as:

$$r(t) = \begin{cases} -\beta_1 \cdot \sum_{t'=t-\Delta t}^t EC(t') - \beta_2 \cdot \sum_{t'=t-\Delta t}^t \left( \sum_{i=1}^n c_{penalty,i}(t') \right), & O_{cc} = 1 \\ 0, & O_{cc} = 0 \end{cases} \quad (3)$$

In Eq. (3),  $\Delta t$  is the control interval; the first term is the energy cost of the HVAC system, where  $EC(t')$  represents the total energy consumption of HVAC system in kWh.  $O_{cc}$  represents the occupancy, with a value of 1 for occupancy between 8 a.m. and 7 p.m., and a value of 0 for other times. The coefficient  $\beta$  is used to weight the energy and thermal comfort terms. When the value of  $\beta$  is set to 1, the aim is to save as much energy as possible, but neglects the need for the thermal comfort of occupants. On the other hand, when the value of  $\beta$  is set to 0, the agent only focuses on maintaining the indoor temperature at the desired setpoint. The control objective is to provide balance between these two extremes. The determination process of the coefficient  $\beta$  is discussed in detail in Section 4.3. The second term is the penalty for indoor temperature deviation, which is calculated as follows:

$$c_{penalty,i}(t') = \begin{cases} 1, & \text{for } T_{in,i}(t') < T_{sp} - T_{th} \text{ or } T_{in,i}(t') > T_{sp} + T_{th} \\ 0, & \text{else wise} \end{cases} \quad (4)$$

In Eq. (4),  $T_{in,i}(t')$  represents the indoor air temperature of zone  $i$  at time  $t'$ .  $T_{sp}$  represents the setpoint temperature.  $T_{th}$  is a small threshold value. Since it is not always possible to keep the indoor temperature at an exact setpoint, the threshold allows for some deviation in the indoor temperature. If the magnitude of the indoor temperature deviation is less than  $T_{th}$ , the temperature deviation is not counted.

Another undefined element of MDP is the state transition probability  $p$ , which refers to the probability of transitioning to a certain next state after taking an action. With a known state transition probability, the MDP is completely observable, and the cumulative reward can be analytically solved using model-based dynamic programming. However, it is quite challenging to establish an accurate thermal dynamics model for the HVAC control problem. The heat transfer within the building is influenced by the material coefficients of various building components. Additionally, the indoor temperature is also impacted by unpredictable external factors like outdoor temperature, solar irradiance, and wind, which necessitates additional modeling and computational work. Therefore, the model-based method is not a simple task for optimizing HVAC systems. Based on the above considerations, this study utilizes model-free DRL methods for multi-zone building HVAC control problems. The model-free DRL methods can continuously interact with the environment and receive feedback without requiring any knowledge of the environment or the state transitions in advance.

### 3.3. DRL-based control method for multi-zone HVAC system

#### 3.3.1. A brief review of deep reinforcement learning methods

The combination of RL and DNN is called the deep RL (DRL) method, which can be roughly divided into two categories: the value-based RL method and the policy-based RL method. The structures of the two types of DRL methods are shown in Fig. 2.

The Deep Q-Network (DQN) is a representative of the value-based DRL method. The input is the current state, and the output is the Q-

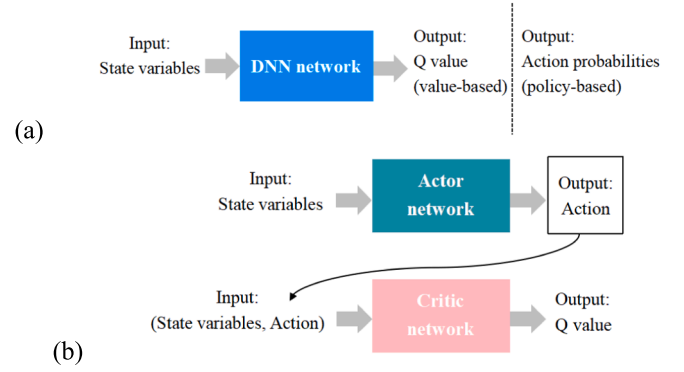


Fig. 2. (a) DQN and (b) DPG structure for function approximation in RL.

value for each action. The DQN algorithm consists of two neural networks: the target network and the behavior network. The target network serves as a reference, similar to the ground truth in supervised learning. Both networks are initialized with the same parameters and structure. During the training, the behavior network updates faster than the target network. The loss function of DQN is the mean squared error (MSE) between the target Q-value and the behavior Q-value. After calculating the loss function, the parameters of the behavior network will be updated based on its gradient to the loss function. The learning process continues until the outputs of the target network and the behavior network are close, which indicates the convergence of the learning. As a representative of the policy-based DRL method, Deep Policy Gradient (DPG) utilizes a different strategy for control optimization compared to the DQN method. The output is the probability of each action at the current state or policy. The policy refers to the probability of selecting action  $a(t)$  in state  $s(t)$  and is denoted as  $\pi(s, a|\theta)$ .  $\theta$  represents the parameters of the probability function. The objective of the DPG method is to obtain the parameters of the policy  $\pi$ , which can maximize the total reward under the policy  $\pi(s, a|\theta)$  [25], and can be expressed as:

$$\text{maximize } E_{\pi(s,a|\theta)}[\mathcal{R}] \quad (5)$$

#### 3.3.2. DRL algorithm for developing HVAC control strategy

Actor-Critic (AC) is a RL algorithm that consists of two types of neural networks: the actor network and the critic network [35]. Their functions are as follows: the input of the actor network is the current state parameters, and it makes decisions based on the observation of the environment and the current policy. The input of the critic network is the state-action pair, which is the current state plus the action generated by the actor network, and the output is the Q-value of the state-action pair. The Q-value is further used to update the parameters of the actor network. The loss function of the actor network is designed to maximize the Q-value under the current policy, following the optimization logic of the DPG method. The loss function of the critic network is the Mean Squared Error (MSE) of the Q-value, following the optimization logic of the DQN method. Therefore, the function of the actor network is to provide output actions, while the critic network is to evaluate the selected actions. In [36], the authors conducted an assessment of various model-free deep reinforcement learning (DRL) algorithms for continuous HVAC control. We take inspiration from this work and investigate the performance of Proximal Policy Optimization (PPO), a variant of the actor-critic paradigm [37]. We selected PPO based on the following considerations. Our HVAC control problem has a continuous action space, for which PPO is well-suited to manage both continuous state space and action spaces. In this study, we focus more on the impact of introducing multi-source transfer learning on RL algorithms, and how to select appropriate source domains to construct an optimal multi-source transfer set. Therefore, we chose PPO to verify the effectiveness of transfer learning on RL algorithms.

The architecture of the DRL-based HVAC control framework is

illustrated in Fig. 3. The process of applying the DRL algorithm to the HVAC control problem involves the following steps. Firstly, the actor network and critic network are randomly initialized. In each iteration, the system state is initialized first, and the control action for the HVAC system is chosen based on the current actor network. In this study, the control action represents the energy supply of each zone. Then, the selected action is implemented in the environment, and the received reward and the subsequent state are observed. The tuples  $(s(t), a(t), r(t), s(t + 1))$  are stored in an experience replay buffer. Once a sufficient number of tuples is collected, a batch of experience tuples is randomly selected to update the actor network and the behavior network. The random selection prevents the high correlation of learning experiences, which maintains the independent and identically distributed assumption during the learning process. The temporal difference (TD-error) error is utilized to update the Q-value, with the target Q-value being the sum of the current reward and the discounted Q-value of the target critic network for the subsequent control interval. Once the loss function is calculated, the behavior critic network updates the parameters  $\theta^Q$  based on the gradient. The target critic network and target actor network update the parameters at a slower rate compared to the behavior network to enhance the stability of learning.

### 3.4. Construction of the optimal multi-source transfer set

There are four commonly-used forms of transfer learning methods [11], including instance transfer, feature transfer, relation transfer, and model transfer (also known as parameter transfer). Since this study aims to learn the knowledge from data in similar buildings, model transfer is a more suitable choice. The main idea of model transfer is to build a pre-trained model using source task data, then take the parameters of the pre-trained model as the initialization parameters of the target task model, and finally fine-tune the model parameters using the data from the target task to make the model adapt to the target task better. Model transfer typically has two forms: keeping the parameters of shallow layers unchanged while retraining the parameters of the deep layers or fine-tuning all network parameters. Although the latter form may lead to a decrease in training efficiency, it minimizes error propagation in

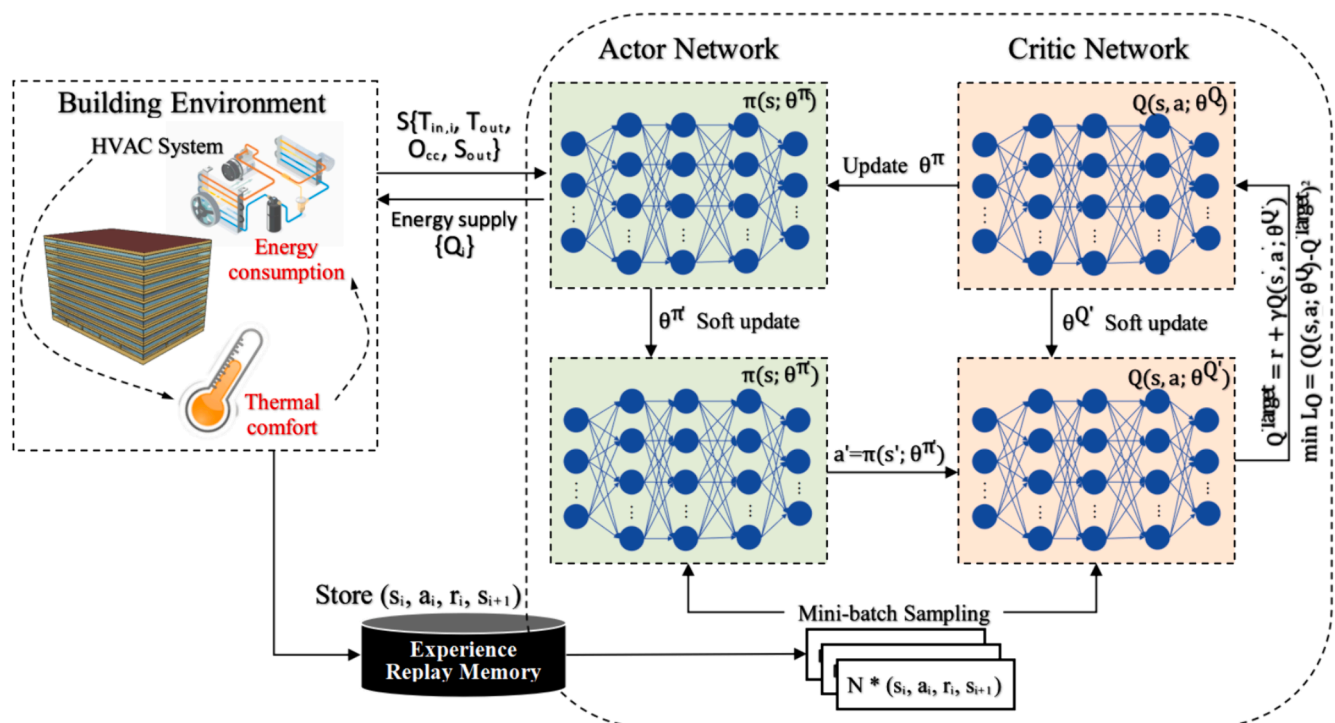
source domains, enhances the safety of scheduling decisions, and yields better performance [38]. Therefore, this study chooses to fine-tune all network parameters.

The typical procedures involve training DRL controllers on a set of sources and then transferring the well-trained parameters to the target domain. As each source DRL controller contributes differently on the transfer task, it is essential to assess the transfer performance of each source domain in order to select valuable sources. Therefore, this study introduces an effective source evaluation framework to quantify the transfer performance of source domains and construct a subset of source domains for multi-source transfer.

Our setup consists of a single target domain and multiple source domains, where the target domain is denoted as  $T$  and the source domain set is represented as  $S=\{S_1, S_2, \dots, S_m\}$ . Our objective is to quantify the transfer performance of each source domain  $S_j$  to the target domain  $T$ . Once the quantification of the contribution of the source domains is completed, we can select source domains with the most significant contribution to construct a source domain subset. The well-trained source controller parameters are then transferred to the target DRL controller to obtain  $C_{T-s_j}$ , representing the transfer of source controller  $S_j$  to the target  $T$ . The corresponding transfer performance of each source domain is stored and compared with the baseline performance to rank the contribution of each source controller.

In this study, we measure the contribution of each source domain from two aspects: energy consumption and temperature deviation, as depicted in Fig. 4. In the non-transfer learning setting, the energy consumption and temperature deviation of the target domain are denoted by  $EC_{base}$  and  $\overline{\Delta T}_{base}$ . After applying transfer learning, the transfer performance of the source domains to the target domain is represented by  $\{EC_{T-S1}, EC_{T-S2}, \dots, EC_{T-Sm}\}$  and  $\{\overline{\Delta T}_{T-S1}, \overline{\Delta T}_{T-S2}, \dots, \overline{\Delta T}_{T-Sm}\}$ . The energy consumption and temperature deviation are first normalized by min-max normalization, which is a widely used pre-processing skill for machine learning [39] (see Eq. (6) and Eq. (7)).

$$EC' = \frac{EC - EC_{min}}{EC_{max} - EC_{min}} \quad (6)$$



**Fig. 3.** The architecture of the HVAC control framework with DRL method.

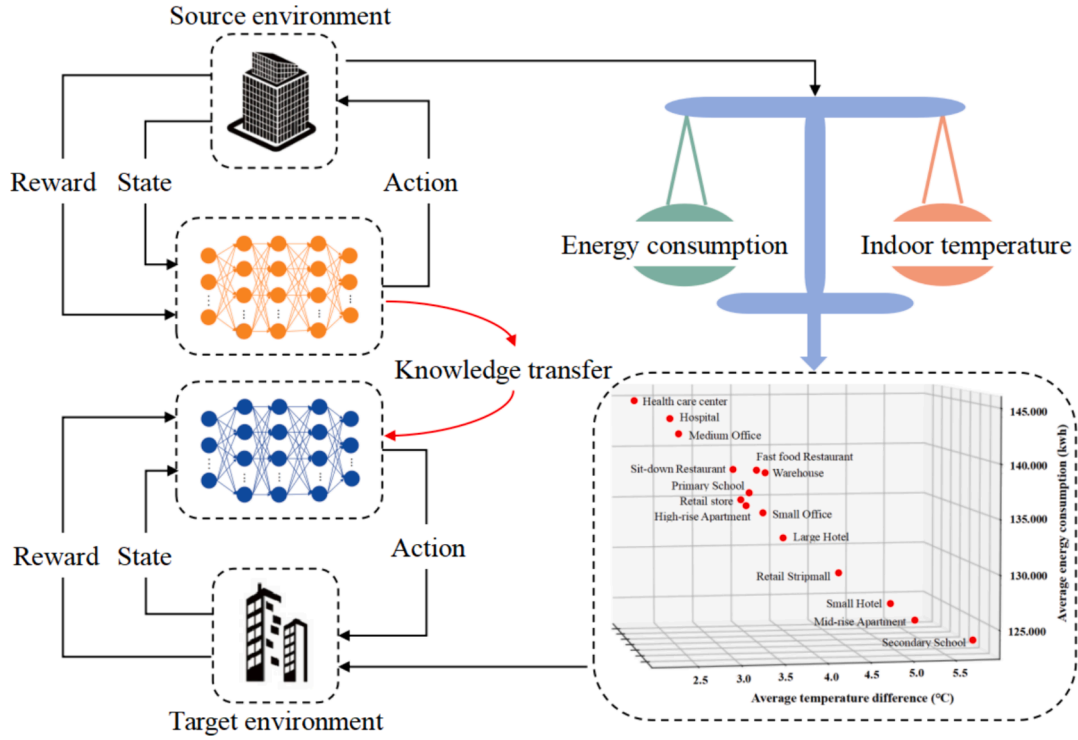


Fig. 4. A framework for evaluating contributions of various source domains.

$$|\Delta T'| = \frac{|\Delta T| - |\Delta T|_{\min}}{|\Delta T|_{\max} - |\Delta T|_{\min}} \quad (7)$$

The contribution of each source is determined by the difference in control performance when trained with and without the source transfer, thus the contribution function of each source domain can be defined as Eq. (8).

$$\begin{aligned} \Delta \text{Score}_{T-sj} &= \text{Score}(C_{sj}, T) - \text{Score}(T) \\ &= m \times (EC'_{T-sj} - EC'_{base}) + n \times (|\Delta T'|_{T-sj} - |\Delta T'|_{base}) \end{aligned} \quad (8)$$

where  $\text{Score}(C_{sj}, T)$  represents the transfer performance of a set of source domains to the target domain, and  $\text{Score}(T)$  represents the control performance without transfer learning.  $m$  and  $n$  represent the weights of two indicators in the scoring function. The contribution of each source domain is sorted to select source domains with higher scores and form the optimal source domain set for multi-source transfer learning.

### 3.5. Proposed MTL-DRL framework of building HVAC control

This section introduces the proposed multi-source transfer DRL-based control method, which selects multiple source domains and transfers the well pre-trained source DRL parameters to the target DRL controller sequentially. The framework mainly consists of the following four steps. (1) The source DRL parameters are randomly initialized, and then pre-trained within the source building until they achieve convergence to an optimal strategy, saving the well-trained source DRL parameters. (2) Source domains with better transfer contributions to the target task are selected to form the optimal source domain set for multi-source transfer learning. (3) The first few layers of well-pretrained source DRL parameters in the optimal source domain set are transferred to the target DRL controller in sequence. (4) The target DRL controller is obtained using the sample data of the target building after the multi-source transfer process.

When the number of thermal zones in the source building is the same

as the target building, the first few layers of parameters of the actor network and critic network of the target DRL controller are directly transferred from the pre-trained source DRL controller. We assume that two appropriate source domains are selected for multi-source transfer learning after the contribution analysis of source domains. Taking the actor network as an example, the parameters of the DRL controller for source building 1 are  $\omega_s^1 = [\omega_{s1}^1, \omega_{s2}^1, \omega_{s3}^1, \omega_{s4}^1]$  and  $b_s^1 = [b_{s1}^1, b_{s2}^1, b_{s3}^1, b_{s4}^1]$ . The parameters of the DRL controller for source building 2 are  $\omega_s^2 = [\omega_{s1}^2, \omega_{s2}^2, \omega_{s3}^2, \omega_{s4}^2]$  and  $b_s^2 = [b_{s1}^2, b_{s2}^2, b_{s3}^2, b_{s4}^2]$ . The target DRL parameters are  $\omega_T = [\omega_{T1}, \omega_{T2}, \omega_{T3}, \omega_{T4}]$  and  $b_T = [b_{T1}, b_{T2}, b_{T3}, b_{T4}]$ . After transferring the first two layers from the first source domain for the first time, the target DRL parameters can be expressed as:

$$\begin{cases} Y_{s1}^1 = \text{Relu}(\omega_{s1}^1 \bullet X + b_{s1}^1) \\ Y_{s2}^1 = \text{Relu}(\omega_{s2}^1 \bullet Y_{s1}^1 + b_{s2}^1) \\ Y_{T3}^1 = \text{Relu}(\omega_{T3}^1 \bullet Y_{s2}^1 + b_{T3}^1) \\ Y_{T4}^1 = \text{Tanh}(\omega_{T4}^1 \bullet Y_{T3}^1 + b_{T4}^1) \end{cases} \quad (9)$$

The well-trained target DRL controller after the first transfer process is preserved and the second source domain is then utilized to further update the network parameters. After transferring the first two layers from the second source domain, the target DRL parameters can be expressed as:

$$\begin{cases} Y_{s1}^2 = \text{Relu}(\omega_{s1}^2 \bullet X + b_{s1}^2) \\ Y_{s2}^2 = \text{Relu}(\omega_{s2}^2 \bullet Y_{s1}^2 + b_{s2}^2) \\ Y_{T3}^2 = \text{Relu}(\omega_{T3}^2 \bullet Y_{s2}^2 + b_{T3}^2) \\ Y_{T4}^2 = \text{Tanh}(\omega_{T4}^2 \bullet Y_{T3}^2 + b_{T4}^2) \end{cases} \quad (10)$$

where  $X$  represents the input and  $Y$  represents the output of each layer.

Relu is chosen as the activation function for all hidden layers, and Tanh is chosen as the activation function for the output layer.

Upon achieving convergence of the reward, the well-trained source controller is preserved to construct the optimal source domain set. At the same time, a DRL network which is similar to the source controller is built as the target controller. The well-trained source parameters from the optimal source domain set are then transferred to the target controller sequentially, avoiding the need of training the target controller from scratch. This multi-source transfer process reduces the effects of error propagation from a single source domain and enhances the security and accuracy of the model. In this framework, the target controller is trained by fine-tuning all network parameters. In order to prevent the fast update of network parameters from destroying the original meaningful weight information, the learning rate should not be set too large during fine-tuning. This study not only considers the DRL controller transfer between different weather conditions, but also explores transfer from n-zone source buildings to m-zone target buildings, that is, the source and target buildings having different numbers of thermal zones. Therefore, it is essential to check whether the input dimensions of the pre-trained DRL controller align with those of the target DRL controller before conducting transfer learning. If the dimensions are the same, the weights and biases of the pre-trained model can be transferred directly to the target DRL controller. If they do not match, the first layer will be skipped, and the weights and biases of the remaining layers will be transferred from the pre-trained model to the target DRL controller. The pseudo code for this framework can be seen in Algorithm 1.

**Algorithm 1** Multi-source transfer DRL-based control framework

```

1: Source Domain Training:
2: for each source domain  $d = 1, 2$  do
3:   Select source domain: building type  $b_d$ , weather type  $w_d$ , and city  $c_d$ 
4:   Initialize environment  $env_{source-d}$  with  $(b_d, w_d, c_d)$ 
5:   Train PPO model  $\pi\theta_{source-d}$  on  $env_{source-d}$ 
6: end for
7: Target Domain Fine-tuning:
8:   Select target domain: building type  $b_t$ , weather type  $w_t$ , and city  $c_t$ 
9:   Initialize environment  $env_{target}$  with  $(b_t, w_t, c_t)$ 
10:  for each source domain  $d = 1, 2$  do
11:    Load pretrained model  $\pi\theta_{source-d}$  and transfer the first few layers
12:    Fine-tune all layers of the model  $\pi\theta_{target}$  on  $env_{target}$ 
13:  end for
14: Detailed Fine-tuning Procedure:
15: for each episode  $e = 1, \dots, N$  do
16:   Reset the target environment and obtain initial state  $s_0$ 
17:   for each timestep  $t = 1, \dots, T$  do
18:     Select action  $a_t$  according to policy  $\pi\theta_{target}(s_t)$ 
19:     Execute action  $a_t$  and observe reward  $r_t$  and next state  $s_{t+1}$ 
20:     Store transition  $(s_t, a_t, r_t, s_{t+1})$ 
21:   end for
22:   for each timestep  $t = 1, \dots, T$  do
23:     Compute advantage estimate  $\hat{A}_t$  using the value network  $V\phi$ 
24:     Compute rewards-to-go  $\hat{R}_t$ 
25:   end for
26:   Update the policy  $\pi\theta_{target}$  by maximizing the PPO-Clip objective
27:   Update the value network  $V\phi$  by minimizing the value loss
28: end for

```

### 3.6. Performance evaluation metrics of the proposed MTL-DRL control framework

The following metrics are used to evaluate the performance of MTL-DRL control method in terms of training efficiency and control performance.

**Training efficiency.** The training time of each episode is visualized to reflect the impact of transfer learning on the training efficiency of DRL-based control methods.

**Control performance.** The average and standard deviation of temperature deviations are used to evaluate the performance of the proposed method in controlling indoor temperature.

$$\mu_{T,i} = \frac{\sum_1^L T_{t,i}}{L} \quad (11)$$

$$\sigma_{T,i} = \sqrt{\frac{\sum_1^L (T_{t,i} - \mu_{T,i})^2}{L}} \quad (12)$$

where  $\mu_{T,i}$  and  $\sigma_{T,i}$  represent the average and standard deviation of temperature deviations of each zone.  $T_{t,i}$  is the indoor temperature at each time step of each zone.  $L$  is the number of hours per week during the testing process, which is 168.

The average temperature difference is the difference between the average temperature of each zone and the setpoint temperature.

$$T_{avg} = \frac{\sum_i^N \mu_{T,i}}{N} \quad (13)$$

where  $T_{avg}$  represents the difference between the average temperature of all zones and the setpoint temperature.  $N$  is the number of zones.

Average standard deviation of temperature deviation. This parameter is utilized to evaluate the fluctuation degree in the indoor temperature around the set point temperature. The symbols  $\uparrow$  and  $\downarrow$  denote whether the temperature deviation of each zone is increased or decreased compared with the baseline performance. If it is true, it is marked as  $\uparrow$ ; otherwise, it is  $\downarrow$ . The symbols  $\uparrow\uparrow$  and  $\downarrow\downarrow$  denote if the temperature deviations of each zone are more increased or more decreased.

$$\bar{\sigma} = \frac{\sum_i^N \sigma_{T,i}}{N} \quad (14)$$

where  $\bar{\sigma}$  represents the average standard deviation of temperature deviation across all zones.  $N$  is the number of zones.

**Average energy consumption.** In addition to thermal comfort, the proposed control method also aims to achieve energy saving of HVAC systems.

$$\text{Average energy consumption} = \frac{\sum_1^L EC(t)}{L} \quad (15)$$

where  $EC(t)$  is the energy consumption during the testing process.  $L$  is the number of hours per week during the testing process, which is 168.

## 4. Results and discussion

### 4.1. Experiment settings

To evaluate the proposed MTL-DRL control methods for building HVAC systems, we utilized a BEAR platform, a physics-principled Building Environment for Control and Reinforcement Learning [40]. Within this platform, the RL control agent sends the actions to a physics-principled simulation engine for model simulation. Then, the simulation states and obtained rewards are returned to the RL control agent. This process is repeated until the end of the simulation. The BEAR platform facilitates the incorporation of advanced control algorithms into the simulation environment, making it easier to debug and enhancing scalability. Another key feature of the BEAR lies in its ability to allow users to customize the built environment by defining a variety of building types, weather conditions, and other environmental variables. Therefore, we adopted the reference models developed by the U.S. Department of Energy (DOE), which have been widely applied in research related to building energy efficiency and HVAC control [41–44].

Fig. 5 displays multi-zone building models with various purposes and layouts. The HVAC system operates in all zones, with the indoor temperature set to 22°C in each zone and daily operation from 8:00 a.m. to

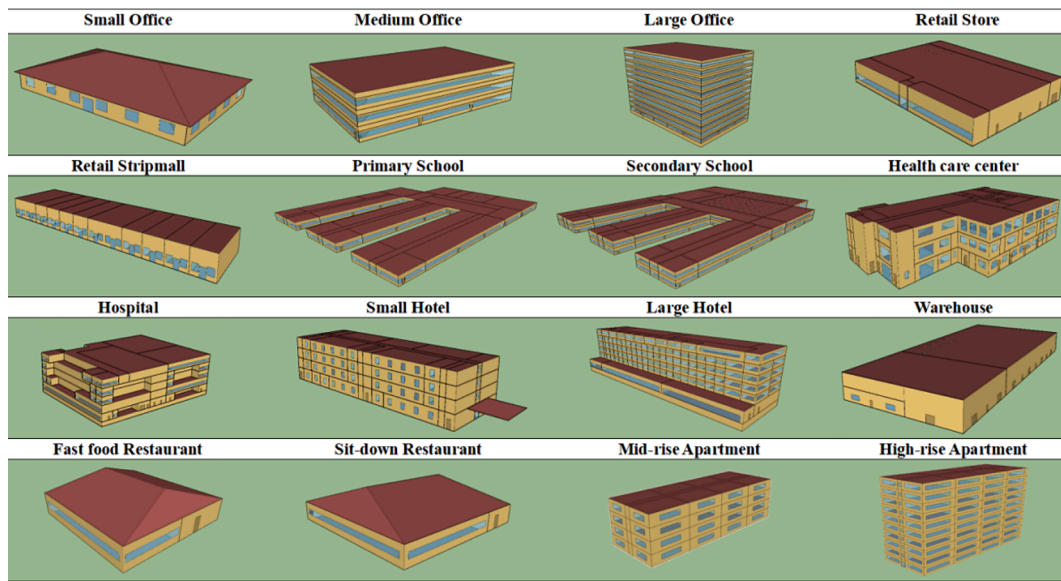


Fig. 5. Multi-zone building models with various purposes and layouts.

7:00p.m. The buildings exhibit varying layouts and numbers of thermal zones, which are more consistent with the diversity of buildings in real-world scenarios. However, this presents more challenges for HVAC control. The Large office building was selected as the target multi-zone building type due to the significant potential for energy conservation

and environmental impact reduction in commercial buildings [45]. The remaining building types serve as training source controllers. Apart from the transferring scenario under different weather conditions, we set the humid climate for all buildings. Detailed explanations are provided in Sections 4.4 and 4.5. The proposed MTL-DRL control method comprises

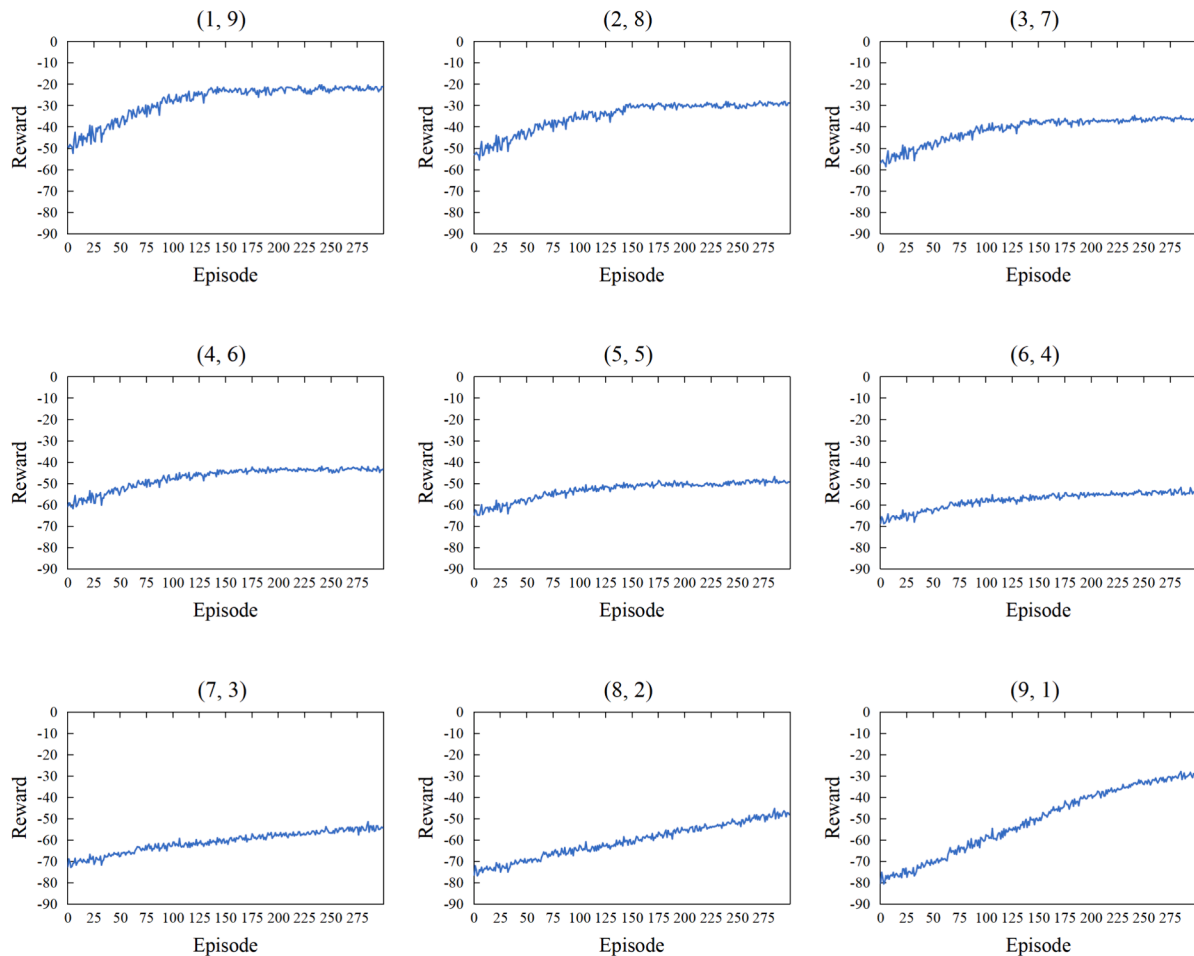


Fig. 6. Cumulative rewards of the weight coefficients under different combinations.

five neural layers. Experimental results revealed that transferring the first two layers provided the best results among all the trials. Subsequently, the performance of the proposed MTL-DRL control method was compared with the DRL baseline method and the single-source transfer learning-deep reinforcement learning (STL-DRL) method to evaluate transfer learning performance from two aspects introduced in Section 3.6. The objective of the proposed MTL-DRL framework is to enhance the training efficiency of the DRL-based control method and reduce energy consumption while maintaining indoor temperatures within an acceptable range.

#### 4.2. Hyper-parameter settings

It is crucial to identify the optimal model hyperparameters in order to achieve the best control performance. Since the weight coefficients of the energy and temperature-related terms significantly impact on convergence and control performance, we focus on the effect of these two key hyperparameters with various configurations. Fig. 6 shows the cumulative rewards of the weight coefficients with different combinations during the DRL training process. It can be seen that the larger the ratios of  $\beta_1$  and  $\beta_2$ , the worse the convergence robustness under the same parameter configurations. When the weight coefficient for temperature violation is less than 6, the convergence of the cumulative reward gradually deteriorates, which is reasonable. A decrease in the penalty coefficient for temperature violation makes it challenging to maintain stable indoor temperatures in multi-zone building HVAC systems. Therefore, it is advisable to assign a relatively larger weight to the temperature term when setting these two key hyperparameters.

Additionally, the control performance is also influenced by the weight coefficients. Fig. 7 depicts the DRL control performance under different combinations of weight coefficients. It can be observed that the ratios of  $\beta_1$  and  $\beta_2$  increase, the energy consumption decreases, but there is a more obvious temperature deviation. Especially when the ratios of weight coefficients change from (3,7) to (4,6), there is a notable increase in temperature deviation. Therefore, it is preferable to choose lower ratios of weight coefficients in order to maintain lower energy consumption while avoiding significant temperature deviations. Based on

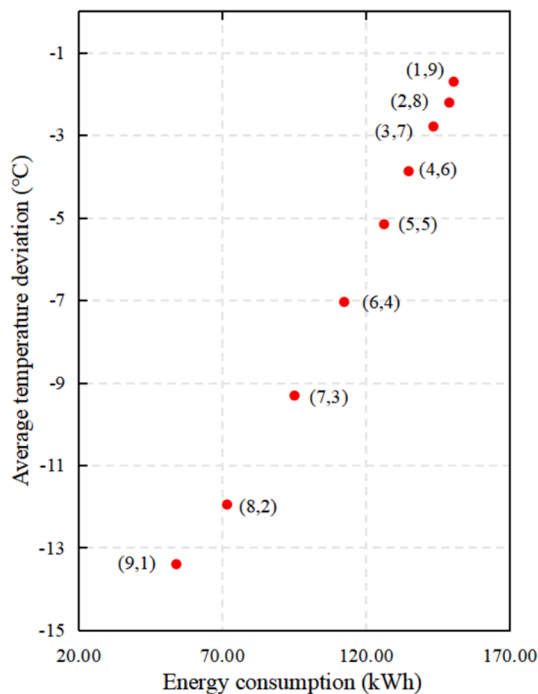


Fig. 7. Control performance of the weight coefficients under different combinations.

the above analysis, the weight coefficients for energy and temperature-related terms are determined as (3,7).

#### 4.3. Results of the optimal source domain set

This section discusses the contribution of different source domains to the target task from computational efficiency, energy consumption, and temperature deviations. Firstly, the computational efficiency of the proposed method is primarily influenced by the number of building zones. Among the 16 sets of building models, the building types with fewer number of thermal zones include Warehouse, Fast food restaurant, and Sit-down restaurant, each consuming about 30 min of training time. The building types with more zones are Health Care Center and High-rise Apartment, with the training time about 45 min each. In single-source transfer experiments, the training time of the target task can be reduced by up to 25.7 % using Warehouse as the source domain, while using Health Care Center as the source domain only results in a 2.9 % reduction in training time, proving the effectiveness of transfer learning in improving training efficiency. Overall, the results demonstrate the effectiveness of transfer learning in improving training efficiency.

Fig. 8 illustrates the impact of various source domains on the energy consumption of the target task. The optimal source domain is the Secondary school, which can effectively reduce the average HVAC system energy consumption by 8.1 %. The next best source domains are the Mid-rise apartment and the Large hotel, which can reduce the average HVAC system energy consumption by 6.8 % and 5.8 %, respectively. The findings also highlight that improper selection of source domains may lead to an increase in HVAC system energy consumption. For example, utilizing the Health care center as the source domain leads to a rise in the average energy consumption of the target control task by 7.2 %.

The contribution of each source domain is then analyzed from the perspective of temperature deviation. Fig. 9 displays the difference between the average temperature of each thermal zone and the set-point temperature before and after using transfer learning. Fig. 10 shows the extreme and mean values of the absolute differences in average temperature of all thermal zones in the target task under different source domains. In the baseline control method, the maximum temperature deviation of about 6°C is observed in Zone 23, with an average temperature deviation of about 4°C across all thermal zones. In contrast to the findings on energy consumption contribution, the utilization of the Secondary School as the source domain leads to significant temperature deviations, with an average temperature deviation of approximately 6°C across all thermal zones. The temperature deviation in Zone 23 even reaches 8°C. More than 80 % of the thermal zones have temperature deviations exceeding 4°C, indicating a failure to reach the designated set-point temperature using single-transfer control strategies. However, utilizing the Health Care Center as the source domain shows the best control of indoor temperature. The results reveal that the maximum temperature deviation in the target task remains below 5°C, with an average temperature deviation of approximately 3°C across all thermal zones.

Fig. 11 provides a more intuitive visualization of the contribution performance of each source domain in terms of energy consumption and temperature deviation. All source domains can be roughly divided into three groups. Group (1) (G1) has low energy consumption but large temperature deviations, and includes Small Hotel, Mid-rise Apartment, and Secondary school. Group (2) (G2) exhibits moderate performance in both energy consumption and temperature deviation control, and includes Sit-down Restaurant, Fast food Restaurant, Warehouse, Primary School, Retail store, High-rise apartment, Small office, Large hotel, and Retail Stripmall. Group (3) (G3) shows high energy consumption but small temperature deviations, and includes Health care center, Hospital, and Medium office. It is clearly necessary to use a comprehensive evaluation index to evaluate the contribution of each source domain to the target control task from both energy consumption and temperature, which is crucial to balance the contributions from both aspects and

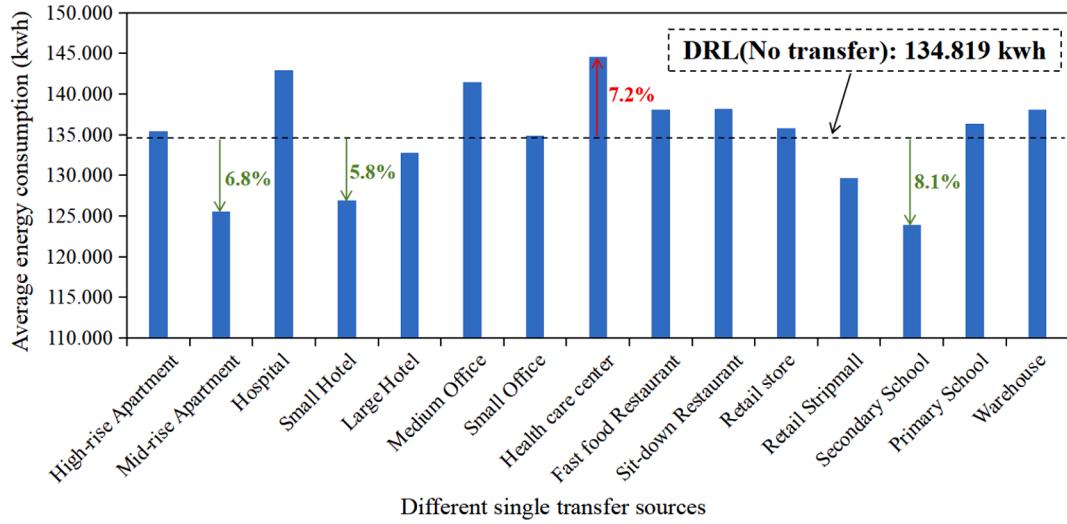


Fig. 8. The impact of different source domains on the energy consumption of the target task.



Fig. 9. The difference between the average temperature of each thermal zone and the set-point temperature.

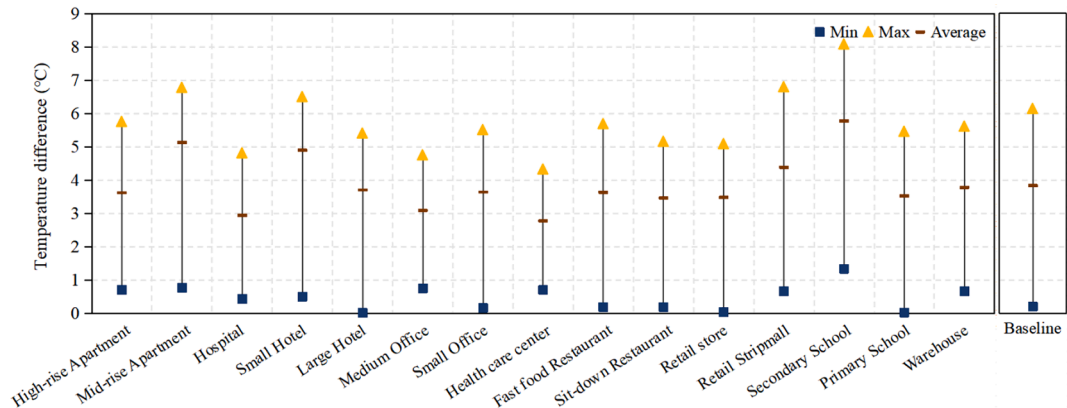


Fig. 10. The distribution of the absolute value of the average temperature deviation for each source.

construct the optimal source domain set.

According to Equations (6) and (7), the average energy consumption and average temperature deviation are firstly normalized. Then the comprehensive scores for each source domain are calculated based on Eq. (8). The weight of each scoring indicator can be adjusted according to the real scenario. In this study, we assigned equal weights of 0.5 to

each indicator. The results shown in Fig. 12 indicate that the Large Hotel and Retail Store are the source domains with the highest scores, both with scores of 0.62. Therefore, the optimal source domain set comprises the Large Hotel and Retail Store. After that, the pre-trained network parameters from the optimal source domain set will be sequentially transferred to the target controller in the multi-source transfer learning

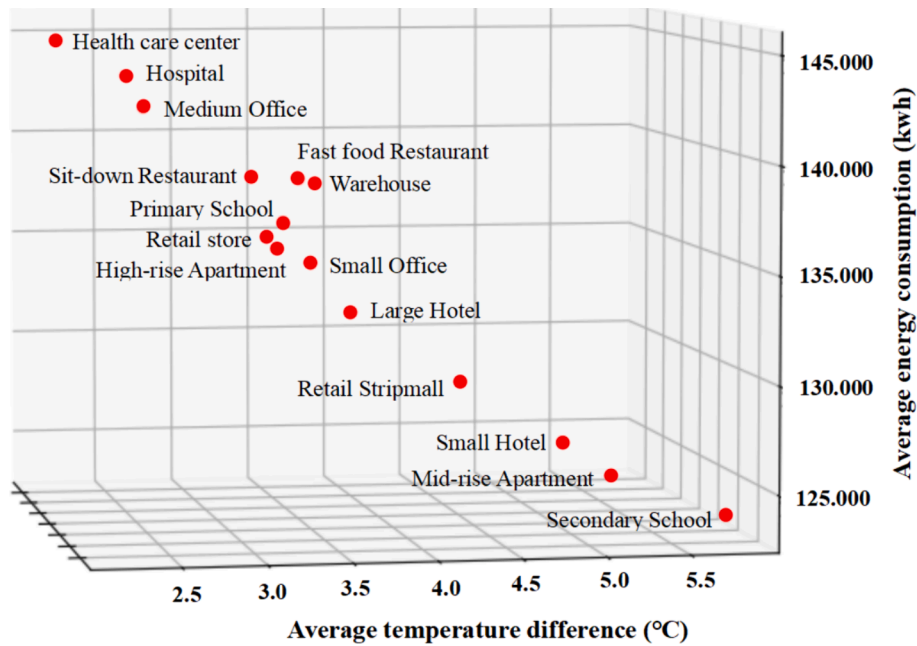


Fig. 11. Visualization of the contribution performance of each source domain.

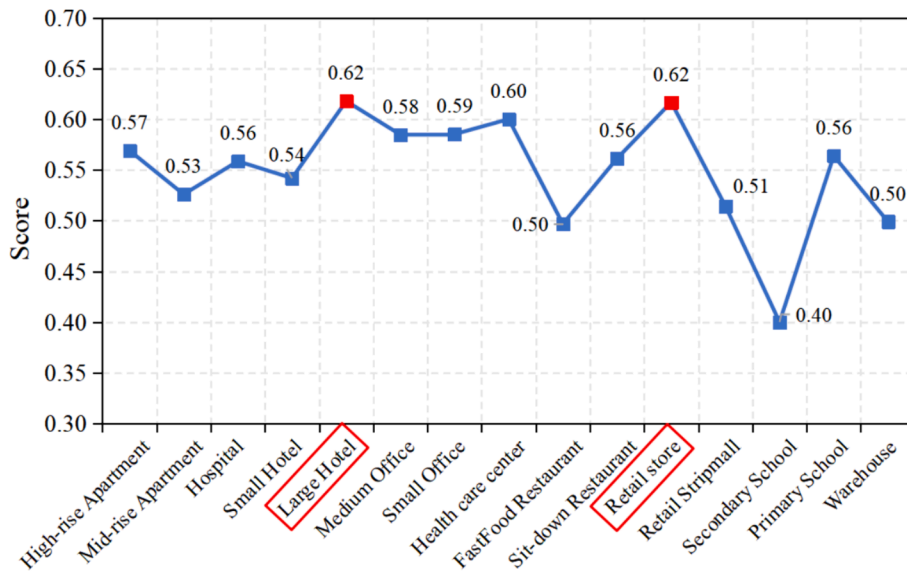


Fig. 12. Comprehensive score of each source domain.

process to enhance the efficiency of the target HVAC control strategy.

#### 4.4. Results of transfer learning performance

##### 4.4.1. Transfer from *n*-zone to *m*-zone in the sequence of Source A and Source B to target

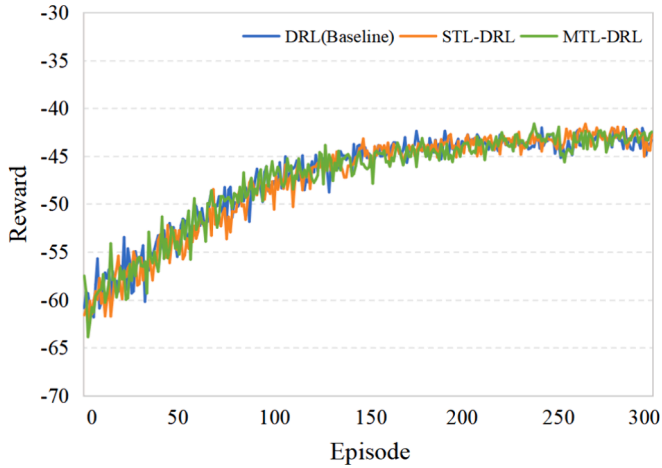
This section evaluates the impact of multi-source transfer on the HVAC system control performance. In the first scenario, the first transfer source domain uses the Retail Store building model, and the second one is the Large Hotel building model to acquire the pre-trained controller. As shown in Fig. 13, the convergence of the three control methods reaches stability as the episode increases.

Fig. 14 illustrates the distribution of training time for each step over 300 episodes for three control methods. It can be observed that the training time for the DRL baseline method is mainly distributed between 5 s and 7 s, while most training time for the STL-DRL and MTL-DRL

methods falls within the range of 4–6 s. The red dashed lines denote the average training times, which are 6.64 s, 5.64 s, and 5.31 s for the DRL, STL-DRL, and MTL-DRL methods, respectively. The training efficiency of the two TL-DRL methods is improved by 15.06 % and 20.03 % compared to the baseline method. These results indicate that the control framework using transfer learning significantly reduces training time compared to the DRL-based control methods trained from scratch.

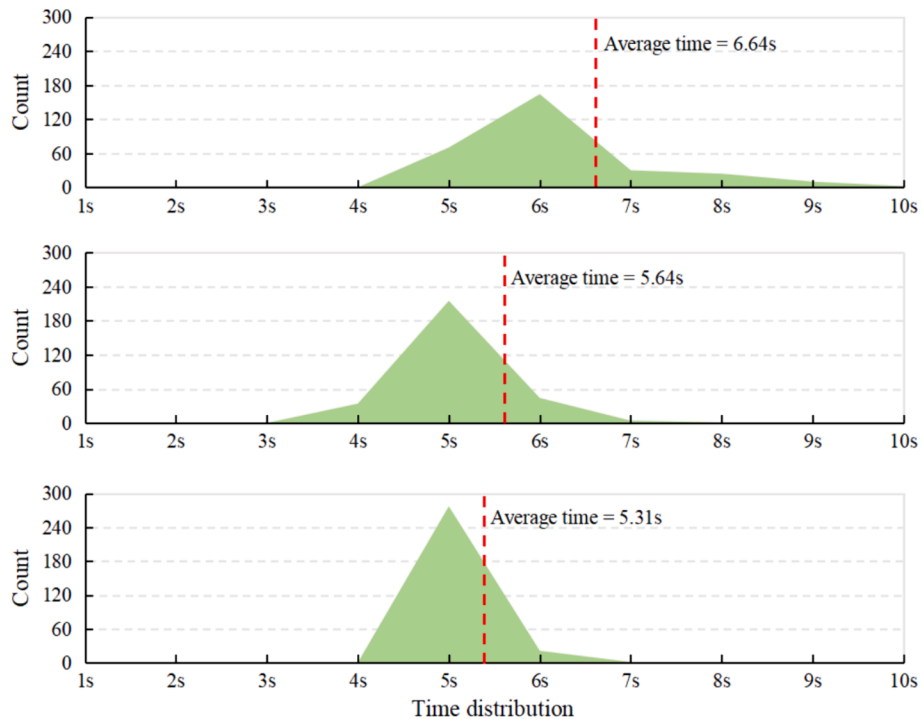
Fig. 15 presents the average energy consumption of three control methods. Specifically, the average energy consumption for the DRL, STL-DRL, and MTL-DRL control methods are 134.820kWh, 135.826kWh, and 132.149kWh, respectively. The average energy consumption using the STL-DRL control method increases by 0.75 %, whereas the MTL-DRL control method reduces average energy consumption by 1.98 %.

Fig. 16 displays the distribution of the average temperature deviation for each thermal zone. Zones 11, 12, 14, and 15 exhibit noticeable



**Fig. 13.** Convergence performance of three control methods in the first transfer scenario using the Retail Store building model as the first source domain and the Large Hotel building model as the second source domain.

temperature deviations regardless of whether the STL-DRL or MTL-DRL control method is utilized, highlighting the challenge of reducing temperature deviations in multi-zone buildings. The average temperature deviations for the DRL, STL-DRL, and MTL-DRL methods are 4.33°C, 4.14°C, and 4.24°C, respectively. Moreover, there is an improvement in temperature deviations in most thermal zones, indicating that incorporating transfer learning into the control framework has the potential to decrease temperature deviations in multi-zone buildings. Table 2 further demonstrates the standard deviation values of the temperature deviations to assess temperature fluctuations in each thermal zone. A comparison between the two TL-DRL control methods reveals that the MTL-DRL control method can reduce the standard deviation of temperature deviations for most thermal zones, giving more stable temperature fluctuations within each thermal zone. In summary, the proposed MTL-DRL control method shows a better transfer learning effect in terms of both training efficiency and control performance.



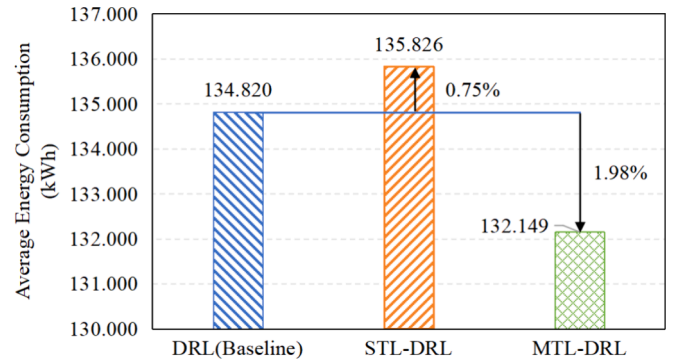
**Fig. 14.** Cumulative count of training time distribution of three control methods.

#### 4.4.2. Transfer from $n$ -zone to $m$ -zone in the sequence of Source B and Source A to target

In the second scenario, we changed the transfer sequence of source domains, where the first transfer source domain uses the Large hotel building model, and the second one uses the Retail store building model to acquire the pre-trained controller. As shown in Fig. 17, the convergence of the three control methods reaches stability as the episode increases.

Fig. 18 shows the training time distribution of each step during 300 episodes for three control methods. The training time of the DRL baseline model is primarily between 5 s and 7 s, while the training times for the STL-DRL and MTL-DRL models range from 4–6 s. The red dashed lines denote the average training times, which are 6.64 s, 5.36 s, and 5.37 s for the DRL, STL-DRL, and MTL-DRL methods, respectively. The training efficiency of the two TL-DRL models is improved by 19.28 % and 19.13 % compared to the baseline model, indicating the effectiveness of transferring pre-trained neural layers in improving training efficiency.

Fig. 19 illustrates the average energy consumption of the three control methods. The average energy consumptions for the DRL, STL-DRL, and MTL-DRL control methods are 134.820 kWh, 132.759 kWh,



**Fig. 15.** The average energy consumption of three control methods.

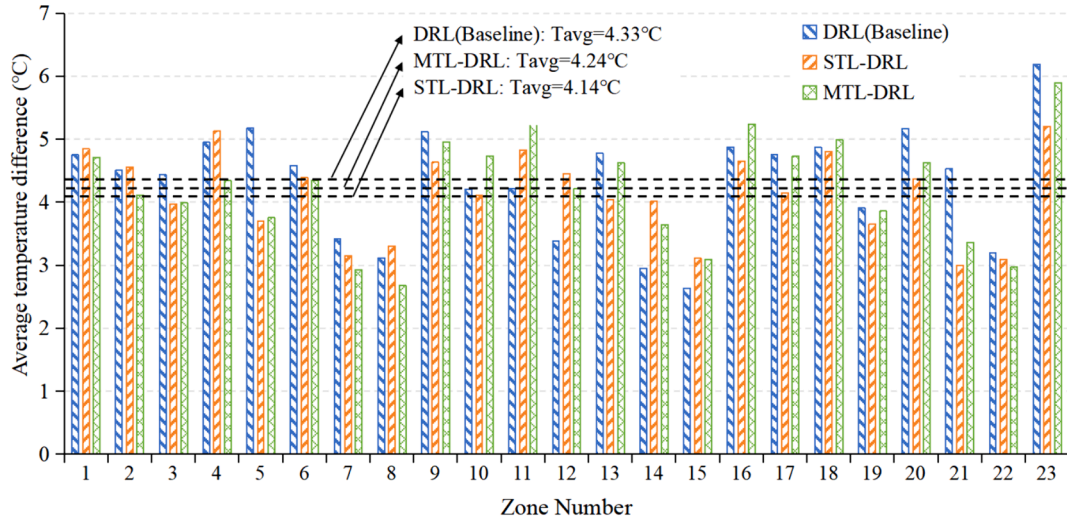


Fig. 16. The distribution of the average temperature deviation for each thermal zone.

Table 2

The change of standard deviation of temperature deviations for each thermal zone.

| Zone Number   | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10   | 11   | 12             |
|---------------|------|------|------|------|------|------|------|------|------|------|------|----------------|
| DRL(Baseline) | 1.51 | 3.64 | 3.77 | 6.61 | 9.61 | 5.92 | 7.77 | 5.81 | 5.81 | 6.37 | 6.58 | 5.27           |
| STL-DRL       | ↑    | ↑    | ↑    | ↑    | ↓    | ↑↑   | ↓↓   | ↓    | ↑    | ↑↑   | ↑↑   | ↑↑             |
| MTL-DRL       | ↓    | ↓    | ↓    | ↓    | ↓↓   | ↑    | ↓    | ↓↓   | ↓    | ↑    | ↑    | ↑              |
| Zone Number   | 13   | 14   | 15   | 16   | 17   | 18   | 19   | 20   | 21   | 22   | 23   | $\bar{\sigma}$ |
| DRL(Baseline) | 8.86 | 3.69 | 4.06 | 7.75 | 7.95 | 9.20 | 7.76 | 9.88 | 8.16 | 5.88 | 9.57 | 7.27           |
| STL-DRL       | ↓    | ↑↑   | ↑    | ↑    | ↑    | ↑    | ↓    | ↓    | ↓    | ↓↓   | ↑    | 7.76           |
| MTL-DRL       | ↓↓   | ↑    | ↑↑   | ↓    | ↓    | ↓    | ↓↓   | ↓↓   | ↓↓   | ↓    | ↓    | 6.77           |

\*The symbols ↑ and ↓ denote whether the temperature deviation of each zone is increased or decreased compared with the baseline performance. If it is true, it is marked as ↑; otherwise, it is ↓. The symbols ↑↑ and ↓↓ denote whether the temperature deviation of each zone is more increased or more decreased.

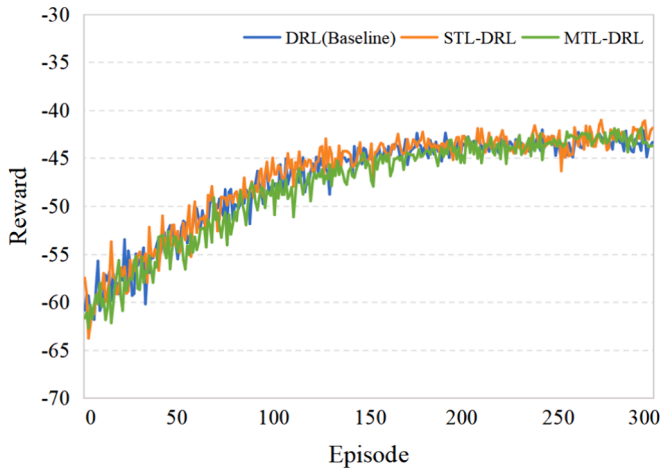


Fig. 17. Convergence performance of three control methods in the second transfer scenario using the Large Hotel as the first source domain and the Retail Store as the second source domain.

and 139.022kWh, respectively. Contrary to the energy consumption analysis in the first transfer scenario, the average energy consumption using the STL-DRL control method reduces by 1.53 %, whereas the MTL-DRL control method increases average energy consumption by 3.12 %.

Fig. 20 shows the distribution of the average temperature deviation for each thermal zone. The average temperature deviations of the DRL, STL-DRL and MTL-DRL models are 4.33°C, 4.25°C and 3.71°C respectively. Except for Zones 11, 12, 14, and 15, using the MTL-DRL control

method reduces the temperature deviation in most thermal zones. The results show that the control method combined with TL is beneficial to reduce the temperature deviation, especially using the MTL-DRL control method, which reduces the overall temperature deviation by 14.32 %. Table 3 also provides the standard deviation of the temperature deviations for each thermal zone. The STL-DRL control method slightly increases the temperature fluctuation in the thermal zone, while the MTL-DRL method reduces the standard deviation of temperature deviation by about 1 °C. Therefore, employing the MTL-DRL control method can enhance the stability of temperature fluctuations in most thermal zones.

The experimental results of the two groups of source domain transfers with reversed order are different. When the Retail Store was used as the first source domain and the Large Hotel as the second source domain, STL-DRL increased average energy consumption by 0.75 % and decreased average temperature deviation by 0.09°C compared to the baseline control method, while MTL-DRL reduced average energy consumption by 1.98 % and average temperature deviation by 0.19°C. In the second scenario where the Large Hotel was utilized as the first source domain and the Retail Store as the second source domain, STL-DRL decreased average energy consumption by 1.53 % and average temperature deviation by 0.08°C compared to the baseline control method. Although MTL-DRL increased average energy consumption by 3.12 %, it decreased average temperature deviation by 0.62°C. These findings illustrate that if only the Large Hotel or Retail Store is chosen for single-source transfer, valuable control experiences contained from the unselected source domain would be wasted. The proposed multi-source transfer framework effectively leverages control experiences from different source domains, achieving better control performance in terms of energy consumption and temperature compared to single-source

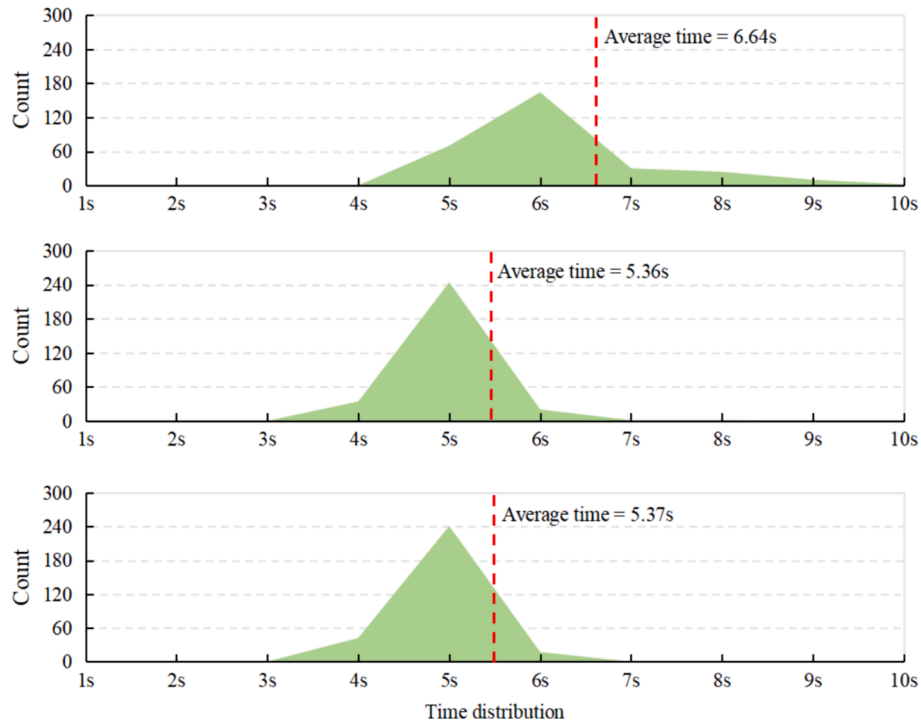


Fig. 18. Cumulative count of training time distribution of three control methods.

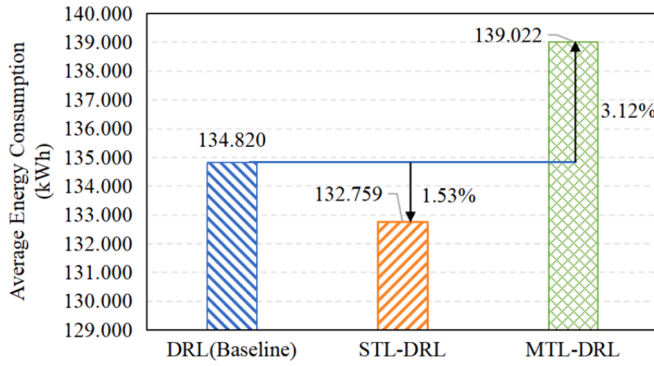


Fig. 19. The average energy consumption of three control methods.

transfer. Additionally, the transfer sequence of the source domain also has an impact on the control performance, which is discussed in detail in Section 4.5.2.

#### 4.5. Discussion

##### 4.5.1. Transfer from *n*-zone to *n*-zone with different weather conditions

This section investigates the potential impact of different weather conditions between the source and target domains on the efficacy of transfer learning in HVAC system control. In addition to differences in model features between the source and target domains, they also have different weather conditions. The initial transfer source domain is established with parameters denoted as Retail store\_Dry, followed by the second transfer source domain designated as Large hotel\_Dry, while the parameters of the target domain are specified as Large office\_Humid. As depicted in Fig. 21, the convergence of the three control methods

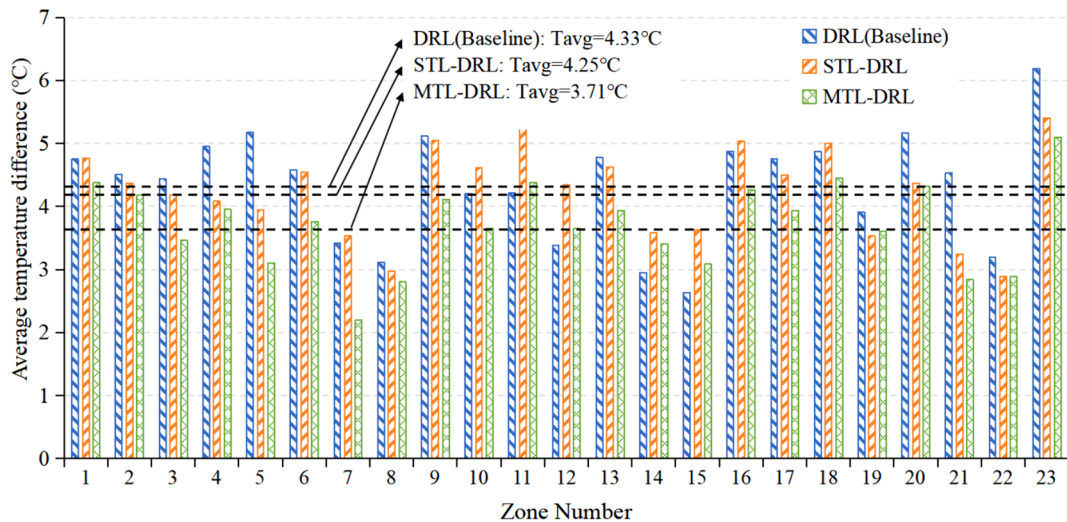


Fig. 20. The distribution of the average temperature deviation for each thermal zone.

**Table 3**

The change of standard deviation of temperature deviations for each thermal zone.

| Zone Number   | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10   | 11   | 12             |
|---------------|------|------|------|------|------|------|------|------|------|------|------|----------------|
| DRL(Baseline) | 1.51 | 3.64 | 3.77 | 6.61 | 9.61 | 5.92 | 7.77 | 5.81 | 5.81 | 6.37 | 6.58 | 5.27           |
| STL-DRL       | ↑    | ↑    | ↑    | ↓↓   | ↓    | ↑    | ↑    | ↑    | ↑    | ↑    | ↑↑   | ↑↑             |
| MTL-DRL       | ↓    | ↓    | ↓    | ↓    | ↓↓   | ↓    | ↓    | ↓    | ↓    | ↓    | ↑    | ↑              |
| Zone Number   | 13   | 14   | 15   | 16   | 17   | 18   | 19   | 20   | 21   | 22   | 23   | $\bar{\sigma}$ |
| DRL(Baseline) | 8.86 | 3.69 | 4.06 | 7.75 | 7.95 | 9.20 | 7.76 | 9.88 | 8.16 | 5.88 | 9.57 | 7.27           |
| STL-DRL       | ↓    | ↑    | ↑↑   | ↑    | ↑    | ↑    | ↓    | ↓    | ↓    | ↓↓   | ↓    | 7.85           |
| MTL-DRL       | ↓↓   | ↑↑   | ↑    | ↓    | ↓    | ↓    | ↓↓   | ↓↓   | ↓↓   | ↓    | ↓↓   | 6.14           |

\*The symbols ↑ and ↓ denote whether the temperature deviation of each zone is increased or decreased compared with the baseline performance. If it is true, it is marked as ↑; otherwise, it is ↓. The symbols ↑↑ and ↓↓ denote whether the temperature deviation of each zone is more increased or more decreased.

reaches stability as the episode increases.

The training time distribution of three control methods is presented in Fig. 22. It can be observed that the training time of the DRL baseline method is mainly distributed between 5 s and 7 s, while the training times of the STL-DRL and MTL-DRL methods generally remain at 4–6 s. The average training time for the DRL, STL-DRL, and MTL-DRL models are 6.64 s, 5.29 s, and 5.28 s, respectively. The training efficiency of the two TL-DRL models is improved by 20.33 % and 20.48 % compared to the baseline model. Similar to the previous analysis of training efficiency, transferring pre-trained neural layers can accelerate the training speed of the DRL control algorithm.

Fig. 23 presents the average energy consumption of three control methods. The average energy consumption of the DRL, STL-DRL, and MTL-DRL control methods is 134.820kWh, 135.573kWh, and 136.743kWh, respectively. Both the STL-DRL and MTL-DRL control methods did not have a positive impact on the energy consumption of the HVAC system, resulting in an increase in average energy consumption by 0.56 % and 1.43 %, respectively.

Fig. 24 reflects the distribution of the average temperature deviation in each thermal zone. Except for Zones 11, 12, 14 and 15, the temperature deviations in the remaining thermal zones are decreased by using the MTL-DRL control methods. The average temperature deviations for the DRL, STL-DRL, and MTL-DRL methods are 4.33°C, 4.20°C, and 3.73°C, respectively. Results indicate that the integration of TL in the control methods is beneficial in reducing temperature deviations, especially when employing the MTL-DRL method, which led to a 13.86 % reduction in average temperature violations for the target task. Table 4 displays the standard deviation of the temperature violations to assess the temperature fluctuation in each thermal zone. Implementation of the STL-DRL control method results in a slight increase in temperature fluctuation, while the MTL-DRL method is capable to reduce

the standard deviation of temperature deviations by almost 1°C, even under different weather conditions between the source and target domains. The experimental results indicate that the MTL-DRL can reduce the deviation between the indoor temperature and the set point temperature of multi-zone buildings at the expense of a slight increase in energy consumption, even if the weather conditions in the source domain and the target domain are different.

#### 4.5.2. The impact of the source domain sequence of transfer learning on the HVAC control performance

The analysis in Section 4.4 demonstrate that the transfer sequence of source domains has an influence on the performance of HVAC control. According to the contribution analysis of each source domain in Section 4.3, we selected source domains with different degrees of improvement in energy consumption and indoor temperature control. We designed three groups of transfer scenarios to explore the impact of the source domain transfer sequence on the control performance. Detailed information is provided in Table 5.

Figs. 25 and 26 show the energy consumption and temperature deviation results of the three groups of transfer scenarios, respectively. The red dashed line represents the energy consumption of the baseline DRL control method. For the first transfer scenario, when Retail Store is used as the second transfer source domain, the energy consumption of the target task increases, but the temperature deviation is reduced compared to when Large hotel is employed as the second transfer source domain. These comparison results are consistent with the performance of Large hotel and Retail store as single source domains in the STL-DRL experiment. In the second transfer scenario, one of the source domains is the Secondary school, which exhibits the best control performance of energy consumption but the worst temperature control effect in the STL-DRL experiment. Whether the Secondary school is designated as the first or second transfer source domain, it did not enhance the temperature control but decreased the average energy consumption. In the third scenario, one of the selected source domains is the Health care center, which exhibits the best results in temperature deviation control but the highest energy consumption. Both experiments in Scenario 3 resulted in increased energy consumption, especially when the Health care center was employed as the second transfer source domain. Nonetheless, both experiments led to a reduction in both the maximum and average temperature deviation.

The outcomes of three experiments highlight the significant influence of the source domain in the second transfer process. The main purpose of the first transfer process is to update the neural network parameters using well-trained parameters, instead of the operation of random initialization. By employing two consecutive transfer learning processes, it becomes possible to effectively achieve better control performance.

## 5. Conclusions and future work

This study proposes a multi-source transfer learning and deep reinforcement learning (MTL-DRL) framework to enhance the building

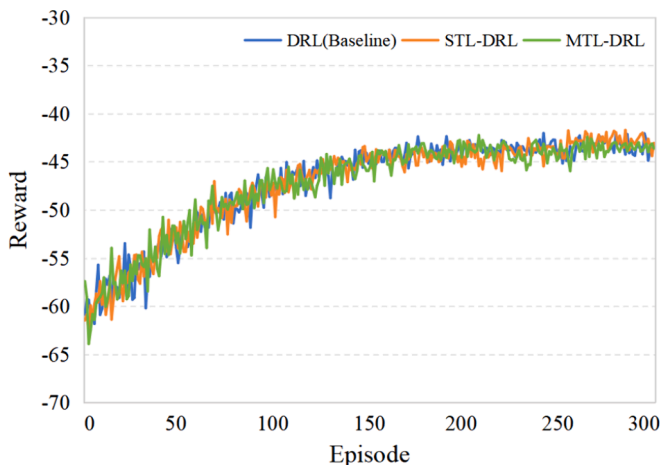


Fig. 21. Convergence performance of three control methods under different weather conditions.

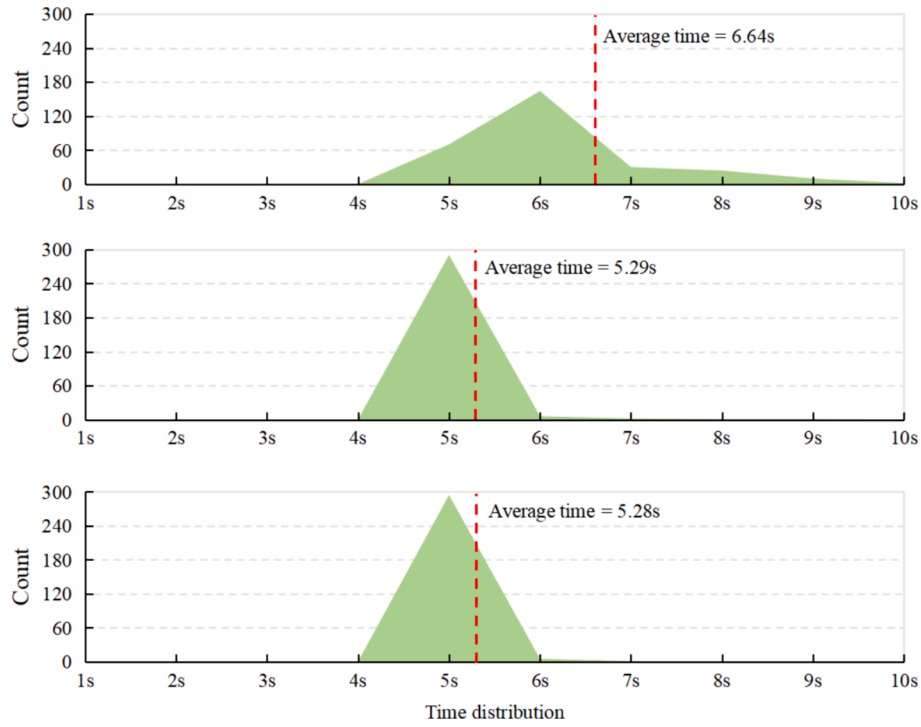


Fig. 22. Cumulative count of training time distribution of three control methods.

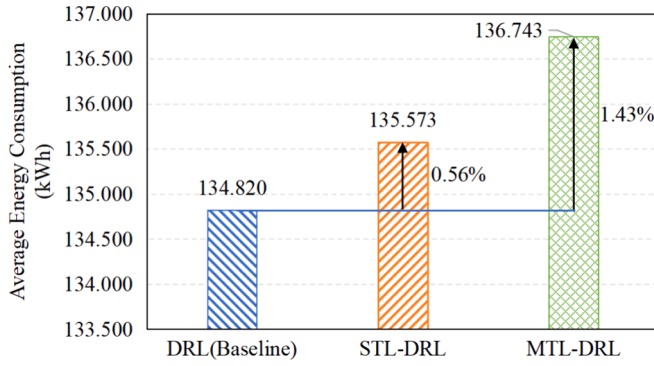


Fig. 23. The average energy consumption of three control methods.

HVAC control by leveraging control experience from different source domains. Firstly, the study quantifies the contribution of each source domain to the target task. Then the comprehensive evaluation of each domain transfer performance based on energy consumption and temperature control is conducted to construct the optimal multi-source transfer set. Subsequently, the pre-trained DRL parameters from various source domains are sequentially transferred to the target DRL controller. The key findings are summarized as follows:

A novel multi-source transfer learning and deep reinforcement learning (MTL-DRL) integrated framework is proposed for multi-zone building HVAC system control. The results indicate that the MTL-DRL framework can improve the training efficiency by up to 20 % compared to DRL baseline models trained from scratch.

This study reveals that the STL-DRL control method reduces average energy consumption by 0.56 %-1.53 %, while the proposed MTL-DRL

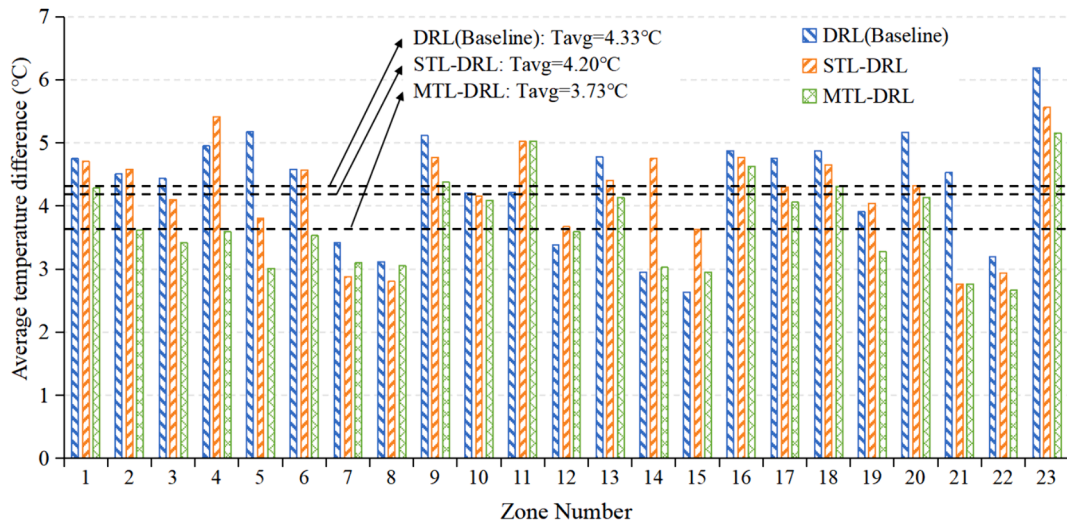


Fig. 24. The distribution of the average temperature deviation for each thermal zone.

**Table 4**

The change of standard deviation of temperature deviations for each thermal zone.

| Zone Number   | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10   | 11   | 12             |
|---------------|------|------|------|------|------|------|------|------|------|------|------|----------------|
| DRL(Baseline) | 1.51 | 3.64 | 3.77 | 6.61 | 9.61 | 5.92 | 7.77 | 5.81 | 5.81 | 6.37 | 6.58 | 5.27           |
| STL-DRL       | ↓    | ↓    | ↑    | ↑    | ↓    | ↑    | ↓    | ↓↓   | ↑↑   | ↑    | ↑↑   | ↑              |
| MTL-DRL       | ↓↓   | ↓↓   | ↓    | ↓    | ↓↓   | ↓    | ↑    | ↓    | ↑    | ↑↑   | ↑    | ↑↑             |
| Zone Number   | 13   | 14   | 15   | 16   | 17   | 18   | 19   | 20   | 21   | 22   | 23   | $\bar{\sigma}$ |
| DRL(Baseline) | 8.86 | 3.69 | 4.06 | 7.75 | 7.95 | 9.20 | 7.76 | 9.88 | 8.16 | 5.88 | 9.57 | 7.27           |
| STL-DRL       | ↓    | ↑↑   | ↑↑   | ↑↑   | ↓    | ↓    | ↑    | ↓    | ↓    | ↓↓   | ↑    | 7.54           |
| MTL-DRL       | ↓↓   | ↑    | ↑    | ↑    | ↓↓   | ↓↓   | ↓    | ↓↓   | ↓↓   | ↓    | ↓    | 6.55           |

\*The symbols ↑ and ↓ denote whether the temperature deviation of each zone is increased or decreased compared with the baseline performance. If it is true, it is marked as ↑; otherwise, it is ↓. The symbols ↑↑ and ↓↓ denote whether the temperature deviation of each zone is more increased or more decreased.

**Table 5**

Detailed information of transfer scenarios.

| Transfer scenario information |                        | Source 1           | Type | Source 2           | Type |
|-------------------------------|------------------------|--------------------|------|--------------------|------|
| Transfer scenario 1           | Ts1                    | Retail Store       | G2   | Large hotel        | G2   |
|                               | Reverse order (Ts1_v1) | Large hotel        | G2   | Retail Store       | G2   |
| Transfer scenario 2           | Ts2                    | Retail Store       | G2   | Secondary school   | G1   |
|                               | Reverse order (Ts2_v1) | Secondary school   | G1   | Retail Store       | G2   |
| Transfer scenario 3           | Ts3                    | Health care center | G3   | Large hotel        | G2   |
|                               | Reverse order (Ts3_v1) | Large hotel        | G2   | Health care center | G3   |

\*According to the analysis of the contribution of each source domain in Section 4.3, all source domains can be divided into three groups. Group (1) (G1) can reduce energy consumption but causes larger temperature deviation. Group (2) (G2) exhibits moderate performance in both energy consumption and temperature deviation control. Group (3) (G3) shows higher energy consumption but reduces the temperature deviation.

control method shows better control performance. Compared to the baseline DRL method, the MTL-DRL control method reduces average energy consumption by about 1.43 %–3.12 %. Furthermore, the MTL-DRL control method reduces the temperature deviation in each thermal zone by up to 14.32 %, with reductions in the average temperature deviations of up to 2°C.

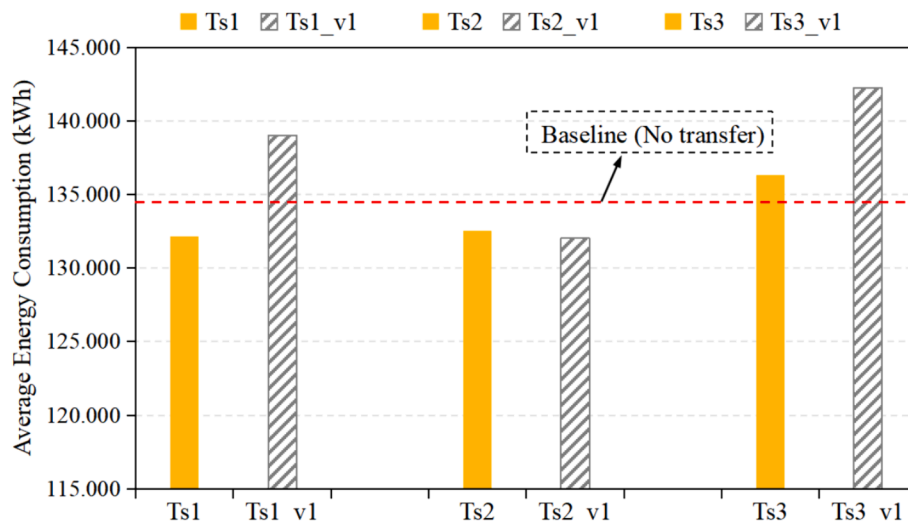
The transfer sequence of source domains significantly affects HVAC control performance. Findings from multi-source transfer experiments show that the source domain in the second transfer process has a more significant impact on the control performance of the target

task. The primary objective of the initial transfer process is to update neural network parameters with well-trained parameters, rather than initializing random parameters. The implementation of two transfer processes effectively enhances the optimization of the control strategy.

The transfer of DRL control strategies within building HVAC systems presents a complex challenge, and this study exists some limitations that can be further improved in future research. The results show that it is difficult to reduce average temperature deviations in some zones. To address this problem, the entire building thermal zone can be divided into different groups. The thermal zones within the same group can be controlled by an independent DRL controller. Then the multi-zone building HVAC system can be decomposed into multiple subsystems to establish multi-agent based DRL control strategies for more efficient HVAC system control. Additionally, transfer learning and *meta-learning* have similarities to a certain extent, but the difference is that the purpose of *meta-learning* is to learn *meta-knowledge* from experiences and facilitate the model to adapt to new tasks quickly [46–49]. Moreover, *meta-learning* typically involves learning higher-level strategies or rules. Therefore, after the optimal multi-source transfer set is constructed, *meta-learning* can be applied to acquire the optimal *meta-knowledge* for similar task learning. Therefore, in our future research, we can explore the different performance between transfer learning and *meta-learning* and integrate these two approaches to enhance the model's generalization.

#### CRedit authorship contribution statement

**Fangli Hou:** Writing – review & editing, Writing – original draft,



**Fig. 25.** The average energy consumption of different transfer scenarios.

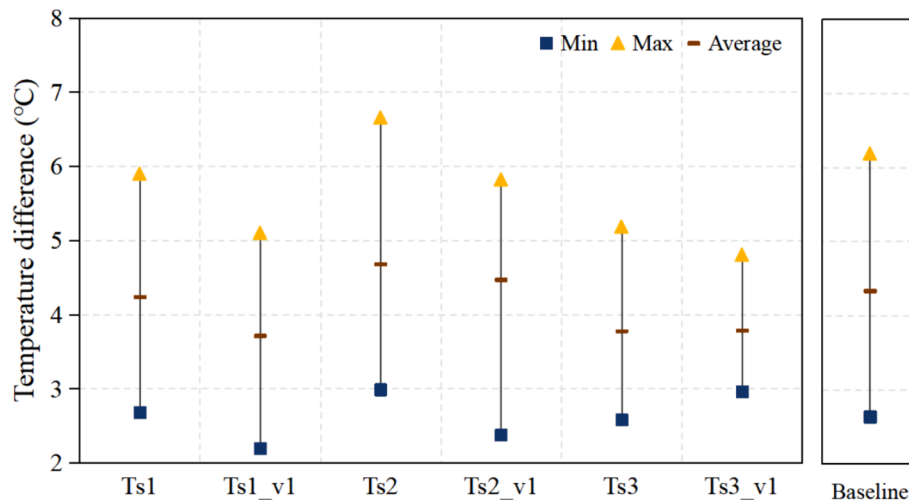


Fig. 26. The distribution of the absolute value of the average temperature deviation of different transfer scenarios.

Validation, Methodology, Formal analysis, Data curation. **Jack C.P. Cheng:** Writing – review & editing, Supervision, Methodology. **Helen H.L. Kwok:** Writing – review & editing, Supervision, Methodology. **Jun Ma:** Writing – review & editing, Supervision, Methodology.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Data availability

Data will be made available on request.

#### Acknowledgment

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.enbuild.2024.114696>.

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